Multifaceted Exploration of Spatial Openness in Rental Housing: A Big Data Analysis in Tokyo's 23 Wards

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Contents

1 Introduction **2**

1.1 Research Background and Motivation 2

1.2 Literature Review and Research Gaps 3

1.2.1 Spatial Openness in Recent Architectural Research 3

1.2.2 Computer Vision and Spatial Analysis 4

1.2.3 Big Data Opportunities and Research Gaps 5

1.3 Primary Research Objectives 5

1.4 Structure of the Thesis 6

2 Methodology **8**

2.1 Overview of Openness 8

2.2 Dataset Characteristics 9

2.2.1 Data Structure and Availability 9

2.2.2 Multi-Stage Data Filtering Process 10

2.3 Spatial Openness Indicators 12

2.4 2D Openness Indicators 14

2.4.1 Visibility Graph Analysis (VGA) Methodology 14

2.4.2 3D Openness Indicators 18

3 Results **21**

3.1 Temporal Trend of Openness Indicators 21

3.1.1 Pairwise Analysis of Temporal Changes for Each Indicator 23
3.2 Geospatial Distribution of Openness Indicators 27
3.3 Correlation Analysis Among Indicators and with Rent and Others 30
3.4 Impact of Openness Indicators on Rental Prices 32
3.5 Visual Validation of Openness Indicators 39
3.5.1 2D Openness Indicators 39
3.5.2 3D Openness Indicators 42
3.5.3 Combined Indicator Analysis 49

4 Conclusions **53**
4.1 Summary of Key Findings 53
4.1.1 Temporal Evolution of Spatial Design 53
4.1.2 Geographic Patterns of Openness 54
4.1.3 Independence of 2D and 3D Openness Measures 54
4.1.4 Market Valuation of Spatial Openness 55
4.2 Future Work 55
4.2.1 VGA Methodology Refinement 55
4.2.2 3D Indicator Robustness 56
4.2.3 Causal Analysis Framework 56
4.2.4 Scale and Generalizability 57

References **59**

Abstract

Understanding spatial openness in residential housing is important for enhancing living quality and guiding architectural design. This study proposes a multifaceted evaluation method of spatial openness in rental housing by leveraging large-scale data from Tokyo's 23 wards. Visibility Graph Analysis (VGA) is applied to floor plan images to assess two-dimensional spatial connectivity. In addition, semantic segmentation is performed on interior photos to capture three-dimensional openness features such as window ratios and ceiling visibility. The study investigates how openness characteristics vary by geographic location and construction year, identifying spatial and temporal patterns. Finally, it explores the relationship between these openness indicators and rental costs, as well as subjective housing preferences, providing insights into how spatial openness influences both market valuation and perceived residential quality.

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Chapter 1

Introduction

1.1 Research Background and Motivation

Understanding spatial openness in residential housing is fundamental to enhancing living quality and guiding architectural design practices. In urban environments, particularly in densely populated cities like Tokyo, the perception of spatial openness significantly impacts residents' psychological well-being, comfort, and overall satisfaction with their living spaces. Traditional architectural evaluation methods have long relied on basic quantitative metrics such as floor area, room count, and ceiling height to assess spatial quality. However, these conventional indicators fail to capture the nuanced spatial qualities that contribute to the subjective experience of openness in residential environments.

The concept of spatial openness extends beyond mere physical dimensions to encompass the complex interplay between visual connectivity, visibility of interior elements, and so on. In the context of rental housing markets, where prospective tenants make decisions based on limited visual information from online platforms, understanding how spatial openness translates through digital representations becomes increasingly important. The proliferation of online rental platforms has created unprecedented opportunities to analyze large-scale housing data, including interior imagery and floor plans, which can provide insights into spatial characteristics that were previously difficult to quantify systematically.

Tokyo's 23 special wards represent a unique data playground for studying spatial openness due to the city's distinctive urban morphology, diverse architectural styles spanning multiple decades, and the comprehensive digitization of rental housing information, which has always been one of the most populared cities and rental market in the world.

1.2 Literature Review and Research Gaps

1.2.1 Spatial Openness in Recent Architectural Research

The study of spatial openness has been approached from various perspectives within architectural and environmental psychology research. Traditional spatial analysis methods have focused on geometric properties and numerical metrics of building attributes. Space syntax theory, developed by Hillier and Hanson, introduced systematic approaches to understanding spatial relationships through graph-based analysis of architectural layouts [1]. Visibility Graph Analysis (VGA), as an extension of space syntax principles, provides a quantitative framework for measuring visual connectivity and spatial integration within architectural spaces.

Recent research has expanded the understanding of spatial openness to encompass cultural and contextual dimensions. AL-Mohannadi et al. demonstrated how spatial configurations in traditional Qatari domestic architecture embody cultural values of privacy and openness through sophisticated spatial arrangements, highlighting the importance of considering cultural context in spatial analysis [2]. This perspective emphasizes that spatial openness is not merely a physical property but a culturally-mediated experience that varies across different architectural traditions.

The methodological approaches to analyzing spatial configurations have also evolved significantly. Kim and Choi introduced Angular VGA and Cellular VGA as innovative extensions of traditional space syntax methods, providing more nuanced ways to analyze human movement behavior in architectural spaces [3]. These methodological advances enable researchers to capture the dynamic aspects of spatial experience that conventional VGA might overlook. Furthermore, recent work by Jahan et al. on wheelchair users' navigation challenges has highlighted the importance of considering diverse mobility perspectives when evaluating spatial accessibility and openness in urban environments [4]. Their framework for "dynamic walkability" underscores how spatial openness must accommodate various user needs and movement capabilities.

The abovementioned previous researches have demonstrated that spatial openness influences occupant behavior, movement patterns, and psychological responses to built environments. Studies in environmental psychology have shown that perceived openness correlates with reduced stress levels, improved cognitive performance, and enhanced overall well-being. However, most existing research has been limited to small-scale case studies or controlled experimental settings, lacking the scale and diversity necessary to understand spatial openness patterns across entire urban regions.

1.2.2 Computer Vision and Spatial Analysis

Recent advances in computer vision and deep learning have opened new possibilities for automated spatial analysis. Semantic segmentation techniques enable the extraction of architectural elements from images, allowing for systematic analysis of interior compositions. Similarly, these deep-learning-based models have been successfully applied to recognize and classify various architectural features in floor plans, including walls, windows, floors, and ceilings. These technological developments create opportunities to process large-scale image datasets that would be impractical to analyze manually.

The integration of 2D image analysis with 3D spatial analysis recently draws more attentions in architectural research. While 2D analysis can capture visual qualities and material compositions, 3D analysis provides insights into spatial relationships and configurational properties. The combination of these approaches offers a more comprehensive understanding of spatial openness than either method alone.

1.2.3 Big Data Opportunities and Research Gaps

The emergence of big data analytics in housing research has revealed new patterns in residential preferences, market dynamics, and architectural trends. Online rental platforms generate vast amounts of data, including property specifications, pricing information, and visual documentation through interior images and floor plans. This data richness enables researchers to investigate questions about spatial quality and market valuation at unprecedented scales, creating opportunities that were previously impossible due to data limitations.

However, the potential for using big data to understand subjective spatial experiences remains largely unexplored, particularly in the context of rental housing markets that traditionally rely on rule-based tabular data without in-depth understanding of spatial qualities. While studies using conventional tabular data can achieve large scale, spatial openness research involving image analysis or human-operated software assessments has been limited to small samples, preventing generalization to broader urban contexts. There have been some explorations in retrieving more abstracted representations of data types like floor plans using machine-driven automated approaches [5], but these methods tend to focus on structural representations with limited interpretation abilities due to the format, such as network-style graphs. Additionally, the tools and methodologies for systematically processing and analyzing large-scale visual and spatial data in housing research are still developing, creating both opportunities and challenges for comprehensive spatial analysis.

1.3 Primary Research Objectives

This research aims to address the identified gaps through the following primary objectives:

1. **Develop a Comprehensive Methodology**: Create a multifaceted evaluation method for spatial openness that combines automated extraction of visual elements from in-

terior images with floorplan semantic segmentation for VGA simulations, enabling large-scale analysis previously limited by manual assessment constraints.

2. **Large-Scale Analysis**: Conduct systematic analysis of spatial openness using large-scale rental housing data from Tokyo's 23 wards.

3. **Temporal and Geographic Investigation**: Examine how spatial openness varies across construction periods and geographic regions within Tokyo's 23 wards.

4. **Feature Dependency Analysis**: Investigate relationships between 2D visual elements and 3D spatial connectivity measures to identify key interactions contributing to spatial openness perception.

5. **Market Correlation Analysis**: Explore the relationships between spatial openness features, impression evaluations and rental costs.

1.4 Structure of the Thesis

This thesis is organized into six main chapters, each addressing specific aspects of spatial openness analysis:

Chapter 1: Introduction provides the research background, literature review, objectives, and methodology overview.

Chapter 2: Methodology presents detailed descriptions of the Lifull dataset, openness quantification methods including interior image-based 3D perceptionanalysis and floor plan-based 2D VGA analysis, and statistical analysis frameworks.

Chapter 3: Results presents the primary findings, including temporal analysis of openness characteristics across construction decades and geographic distribution analysis across Tokyo's 23 wards.

Chapter 4: Conclusions summarizes key findings and outlines future research directions.

The thesis concludes with comprehensive appendices containing detailed technical specifications, additional statistical analyses, and supplementary materials that support the main findings.

Figure 1.1 illustrates the overall framework used in this study, showing the integration of various data sources and analytical approaches to quantify spatial openness.

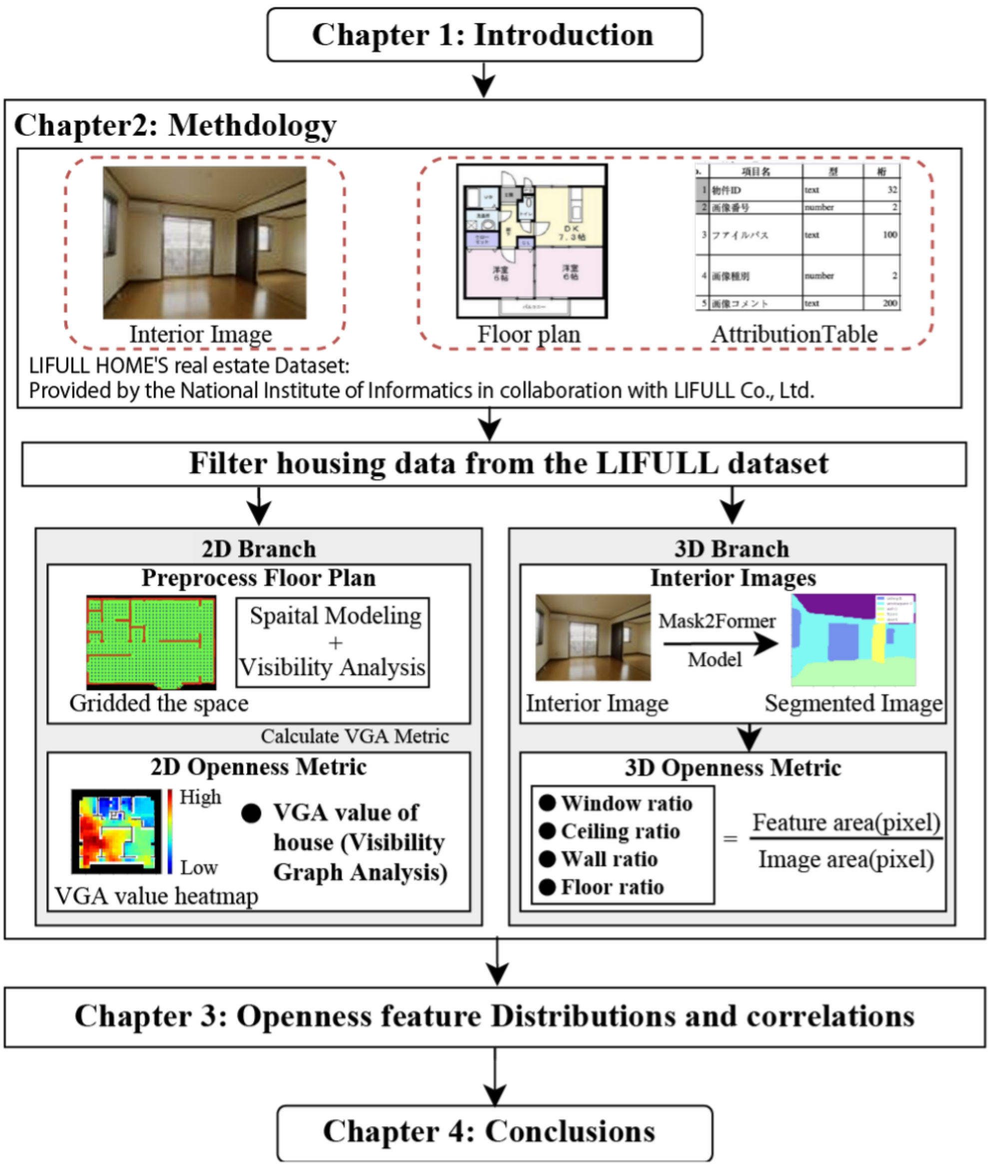


Figure 1.1: Overall framework showing the integration of various data sources and analytical approaches to quantify spatial openness

Chapter 2

Methodology

This chapter presents the data filtering process, dataset characteristics, and methodology for quantifying spatial openness from interior images and floor plan data. The openness indicators' extractions include the windows ratio, ceilings ratio, floors ratio, and walls ratio from interior images, together with tabular based attributes of properties. VGAs relies on the floorplan data. These indicators are used in subsequent temporal, spatial, and correlation analyses.

2.1 Overview of Openness

We consider two types of spatial openness indicators: two-dimensional (2D) openness indicators based on the floor plan analysis extracted from floor plans, and three-dimensional (3D) openness indicators based on the elevational openness extracted from interior images. Each indicator was calculated using computer vision technology from images included in the rental housing big data (LIFULL HOME's dataset). Other information such as construction year, location, and property area is included in the attribute table of the rental properties.

2.2 Dataset Characteristics

The dataset encompasses multiple categories of spatial openness attributes through interior images, floor plan data, and property specifications. Key characteristics include:

2.2.1 Data Structure and Availability

The dataset provides multi-modal information for each property:

- **Interior Images**: High-resolution photographs of living spaces, where we focus on the living room, as shown in Figure 2.1.
- **Floor Plans**: Architectural drawings showing spatial layout and room configurations, as shown in Figure 2.2.
- **Property Specifications**: Detailed attributes including room size, building age, rent prices, and other basic tabular specs, as exampled in Table 2.1. addresses.

Figure 2.1: Example of interior image from the LIFULL dataset.

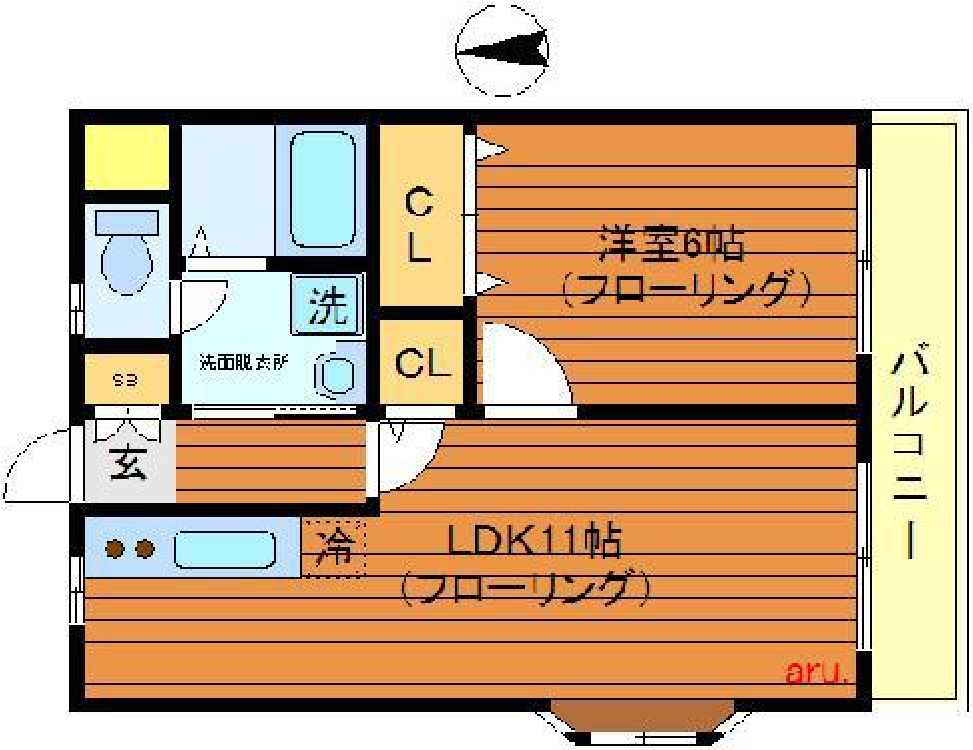


Figure 2.2: Example of floorplan image from the LIFULL dataset.

Table 2.1: Sample of LIFULL dataset structure showing key property attributes

Property ID	**Prefecture**	**City/Ward**	**...**	**Building Area**	**Rent Price**
00026ec12fcea938...	Tokyo	Shibuya	...	51.0	88000
00040e134334f56c...	...	...	...	...	...

2.2.2 Multi-Stage Data Filtering Process

Starting from a raw dataset of over 5 million properties, a systematic multi-stage filtering process is applied to ensure data quality and analytical reliability. The filtering pipeline reduces the dataset to approximately 4,000 final samples suitable for comprehensive openness analysis.

Initial Data Selection

The first stage focuses on selecting properties with high-quality visual data:

- **High-Resolution Floor Plans**: Selection of properties with clear, high-resolution floor plan data only, excluding low-resolution variants to ensure adequate spatial detail for VGA analysis and improved semantic segmentation performance

- **Clear Interior Images**: Properties filtered to include only those with clear living room photographs available, ensuring consistent performance when extracting architectural element ratios through semantic segmentation
- **Geographic Constraint**: Properties limited to Tokyo's 23 wards with accurate location data

Property Specification Requirements

Properties must contain complete tabular specifications including:

- **Location Information**: Accurate geographic coordinates and ward designation within Tokyo's 23 wards
- **Construction Details**: Building construction year, building area, and property type specifications
- **Room Configuration**: Properties must contain room open space area information and at least 1DK room configuration for spatial analysis
- **Essential Attributes**: Complete set of required building-related information for comprehensive analysis

Quality Validation and Final Selection

The final filtering stage ensures analytical validity:

- **VGA Results Validation**: Properties filtered based on legitimate availability of VGA analysis results from floor plan processing

- **Architectural Element Detection**: Verification that semantic segmentation successfully identifies at least one of the key spatial elements (windows, ceilings, floors, or walls) within interior images

- **Multi-Modal Integration**: Final verification that properties contain all required data modalities (high-resolution floor plans, clear interior images, and complete specifications) for comprehensive openness analysis

This systematic filtering process ensures that the final dataset of approximately 4,000 properties maintains high data quality standards while providing sufficient sample size for robust statistical analysis. The entire filtering process is demonstrated in Figure 2.3.

2.3 Spatial Openness Indicators

Spatial openness is typically expressed through the sense of visual connectivity. In residential rental markets, the 3D visual elements displayed in property photographs serve as the primary carriers of spatial impression and openness perception for potential tenants. The concept of openness in residential spaces encompasses several dimensions, with particular emphasis on the following:

- **Visual Connectivity**: The degree to which different areas of a space are visually connected as perceived through interior photographs, which numerically quantified by the VGA analysis.

- **Architectural Elements**: The appearance of windows, ceilings, floors, and walls in photos of the living rooms in the LIFULL HOME's Dataset.

In the context of rental property advertising, these visual elements become the primary means through which spatial qualities are communicated to potential tenants, making their accurate quantification essential for understanding market dynamics.

In order to effectively extract the required openness features, the data filtering and cleaning pipeline ensures data quality and consistency across the dataset. Multiple stages of filtering and validation are applied to maintain analytical reliability. The following sections detail the filtering process for interior images, floor plans, and property specifications, which are also illustrated as shown in Figure 2.3. The details of the methodolgies of how to extract the openness features are demonstrated later after stating the filtering process.

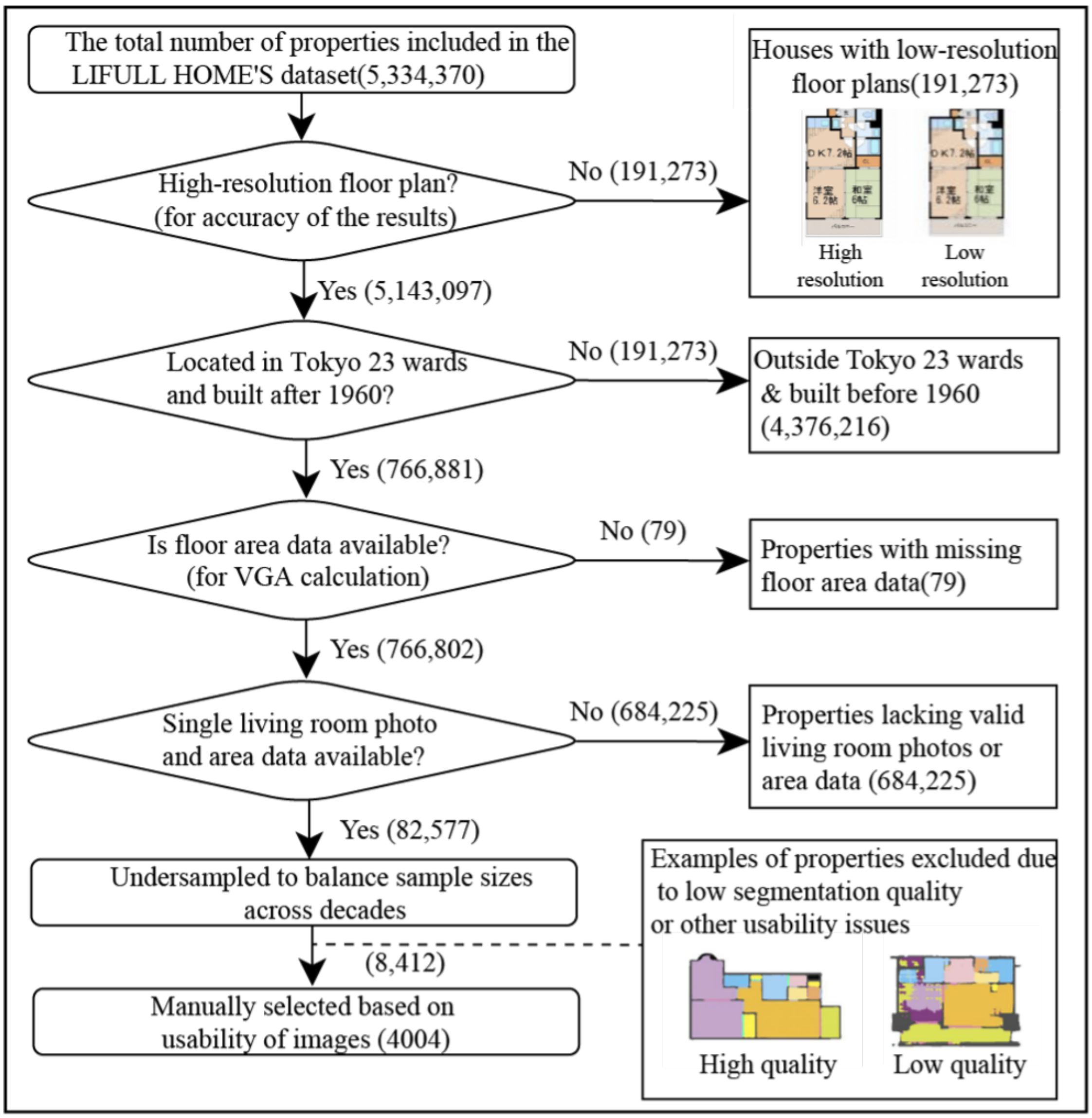


Figure 2.3: Data filtering and preprocessing pipeline showing the systematic approach to ensuring data quality and consistency across interior images, floor plans, and property specifications. The flowchart illustrates the multi-stage filtering process from raw data collection to the final integrated dataset.

2.4 2D Openness Indicators

2.4.1 Visibility Graph Analysis (VGA) Methodology

A key metric for measuring spatial openness is the VGA (Visibility Graph Analysis) value from floor plans, which reflects spatial connectivity. The VGA methodology provides a quantitative approach to assess spatial relationships and visual connections within architectural layouts, complementing the impression-based analysis derived from interior photographs.

Traditionally, VGA analysis requires the transformation of optical floor plans into vectorized data formats that can be processed using specialized third-party software such as depthmapX [6]. This conversion process involves:

- **Digitization**: Converting raster floor plan images into vector-based geometric representations
- **Spatial Calibration**: Ensuring accurate scale and dimensional relationships in the vectorized format
- **Boundary Definition**: Precise delineation of walls, openings, and spatial boundaries for visibility calculations
- **Software Processing**: Utilization of specialized tools like depthmapX for comprehensive spatial analysis

However, this traditional approach presents significant scalability challenges when applied to large-scale datasets. Each floor plan requires individual attention for accurate conversion, and spatial, calibration, showing a bottleneck that limits its applicability to larger data samples due to the extensive human operations involved.

Mathematical Definition of VGA

Corresponding to step 4 in the VGA computational workflow described above, the visibility value quantifies the degree of spatial integration and visual connectivity within a given floor plan layout. For a discretized spatial grid with N points distributed across the open space areas, the visibility graph analysis computes connectivity measures based on line-of-sight relationships between grid points.

Mathematically, the visibility value at each point is defined as:

$$S(i) = \sum_{j=1}^{N} V_{ij} \tag{2.1}$$

where:

- $S(i)$ represents the visibility score at grid point i
- V_{ij} is the visibility indicator function defined as:

$$V_{ij} = \begin{cases} 1 & \text{if point } j \text{ is visible from point } i \\ 0 & \text{otherwise} \end{cases} \tag{2.2}$$

- N is the total number of grid points in the open space areas

The visibility determination between two points follows the isovist principle [7], where point j is considered visible from point i if the straight line connecting them does not intersect any wall boundaries identified through the semantic segmentation process.

This computation generates a VGA heatmap $\mathbf{S} = \{S(1), S(2), \ldots, S(N)\}$ representing spatial connectivity throughout the floor plan. From this heatmap, summary statistics including mean (μ_S), standard deviation (σ_S), and percentiles are extracted as quantitative features characterizing spatial openness for subsequent analysis.

VGA Computation Process

To address traditional VGA limitations, we developed a computational workflow using optical floor plan images directly:

1. **Semantic Segmentation**: DeepLab-V3+ model [8] fine-tuned to classify walls, bedrooms, and living rooms

2. **Boundary Mask Creation**: Generate masks distinguishing walls from open spaces

3. **Physical Grid Generation**: Create uniform grid overlay on open space areas

4. **Isovist-based Visibility Calculation**: Calculate visibility for each grid point using the isovist method

5. **Statistical Extraction**: Extract statistical measures from the resulting heatmap to quantify spatial connectivity

This workflow is demonstrated in Figure 2.4.

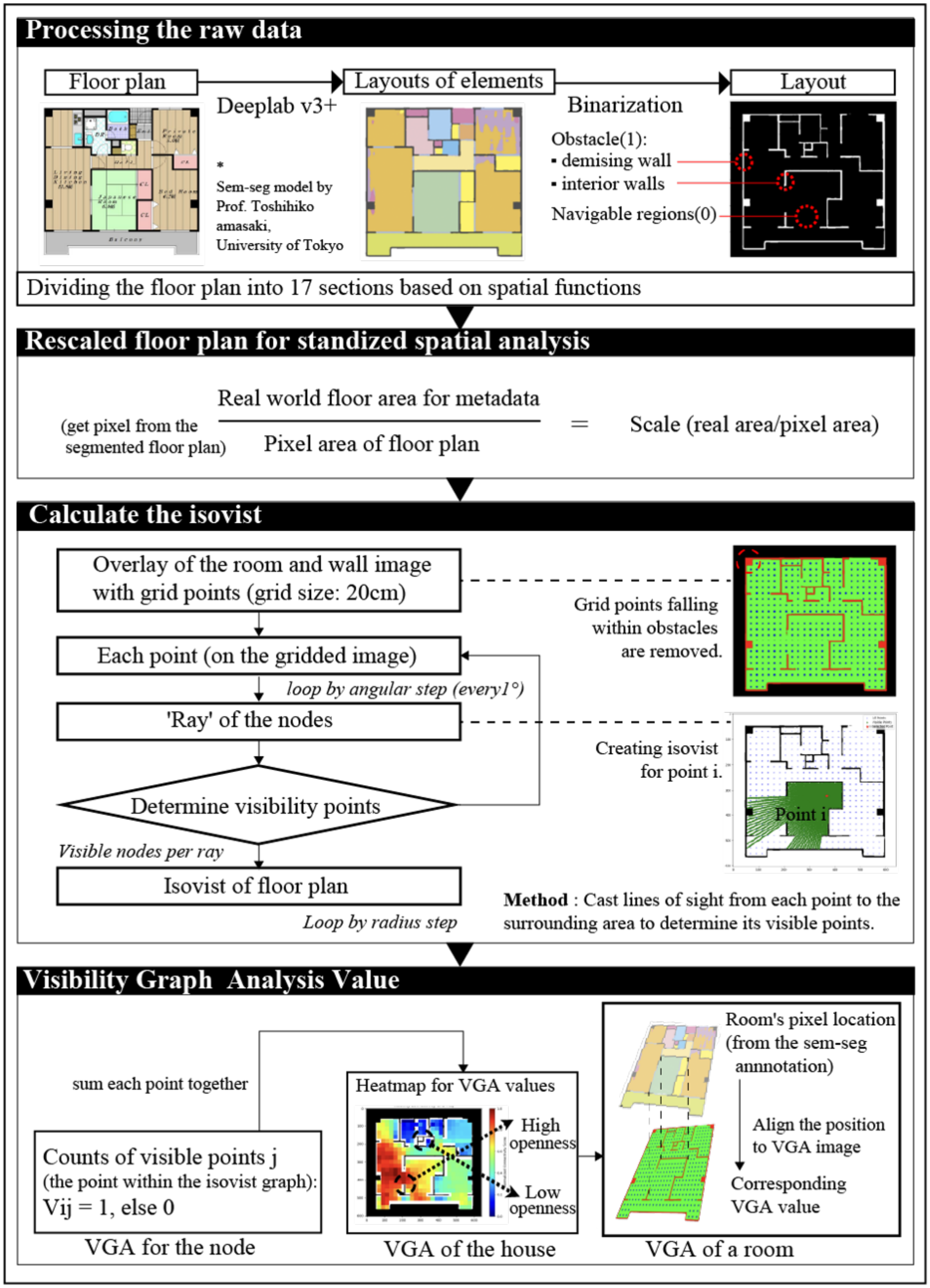


Figure 2.4: VGA computational workflow demonstrating the step-by-step process from semantic segmentation of floor plans to statistical analysis of spatial connectivity.

2.4.2 3D Openness Indicators

Beyond VGA analysis, 3D spatial openness is quantified through the analysis of key architectural elements extracted from interior images. In our studies, such ratios are calculated from the semantic segmentation results of living room images.

- **Windows Ratio**: Proportion of window area relative to total visible area, indicating natural light access and external connectivity
- **Ceilings Ratio**: Ceiling visibility and height perception contributing to vertical spatial openness
- **Floors Ratio**: Floor area visibility affecting spatial grounding and continuity
- **Walls Ratio**: Wall presence and configuration influencing spatial division and flow

These ratios are derived through semantic segmentation techniques applied to interior imagery, providing quantitative measures of spatial composition. However, such extraction may have intrinsic uncertainty due to the photo angles and standing positions collected from the website, but there is hardly any way to eliminate such issues when requesting to use large-scale data from Lifull which is not designed specifically for this study. The methodological details are demonstrated in the following sections.

Segmentation Model

The semantic segmentation employs a Mask2Former pre-trained model [9], which is finetuned on the ADE20K dataset [10] and further optimized for architectural element recognition.

- **Model Selection**: Utilization of state-of-the-art segmentation architectures finetuned on interior image datasets
- **Element Categories**: Classification of pixels into windows, ceilings, floors, walls, and other architectural components

Pixel-wise Classification

The segmentation process generates pixel-wise masks for each architectural element:

1. **Image Preprocessing**: Standardization of input images for consistent processing, including resizing and batching.

2. **Segmentation Execution**: Application of the trained model to generate semantic masks

3. **Post-processing**: recover the original shape and resolutions of the semantic masks to properly calculate the ratios

4. **Refinement Process**: Iterative results by ensuring at least one of the four architectural elements (windows, ceilings, floors, walls) appears in the image.

The above workflow is demonstrated in Figure 2.5.

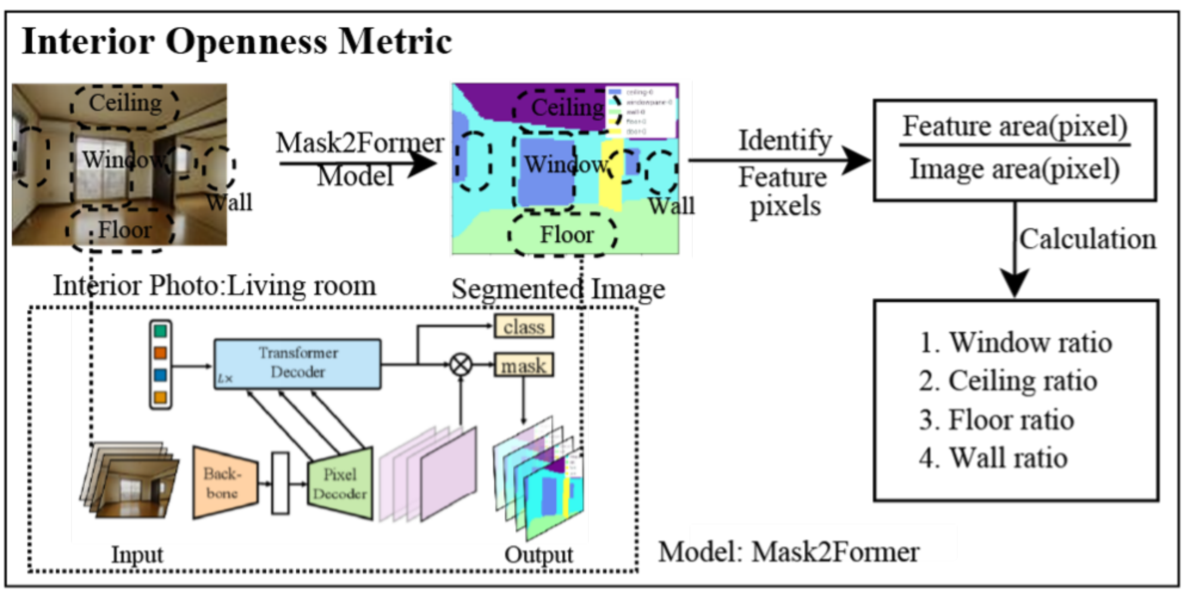


Figure 2.5: Semantic segmentation workflow for architectural element extraction from interior images, showing the process from input image through pixel-wise classification to architectural element ratio calculation.

Ratio Calculation

For each architectural element, the ratio is computed as:

$$\text{Element Ratio} = \frac{\text{Element Pixel Count}}{\text{Total Image Pixels}} \tag{2.3}$$

This provides a normalized measure of each element's presence within the visual field.

Having established the comprehensive data preparation pipeline and feature extraction methodologies, we now transition next chapter for the analytical framework designed to examine the relationship between extracted openness features and human perceptions of spatial openness. This analysis encompasses both individual feature impacts and their collective influence on openness evaluations.

Chapter 3

Results

This chapter presents the analysis results of openness indicators derived from the dataset of 4,004 residential properties in Tokyo's 23 wards, which are selected through the data preparations and filtering process mentioned in the last part of Chapter 2. The findings are organized into three main areas: temporal trends across construction decades, geospatial distribution patterns, and correlation analysis between different indicators, where we hope to unveil some connections between those indicators and the outside factors to strengthen the understanding of how these indicators evolve through time and geographic spaces, and how they correlate with each other.

3.1 Temporal Trend of Openness Indicators

The impact of construction decade on the openness indicators such as mean value of visibility heatmap in whole property and livining room is shown in Figure 3.1. To examine temporal trends, one-way ANOVA (analysis of variance) was conducted for each indicator between each decade group. After confirming the existence of the changes by significance, Tukey's HSD test was applied for post-hoc comparisons to check the difference between each of the decades to understand how much they stand out compared to the others, where significant differences ($p < 0.05$) were found.

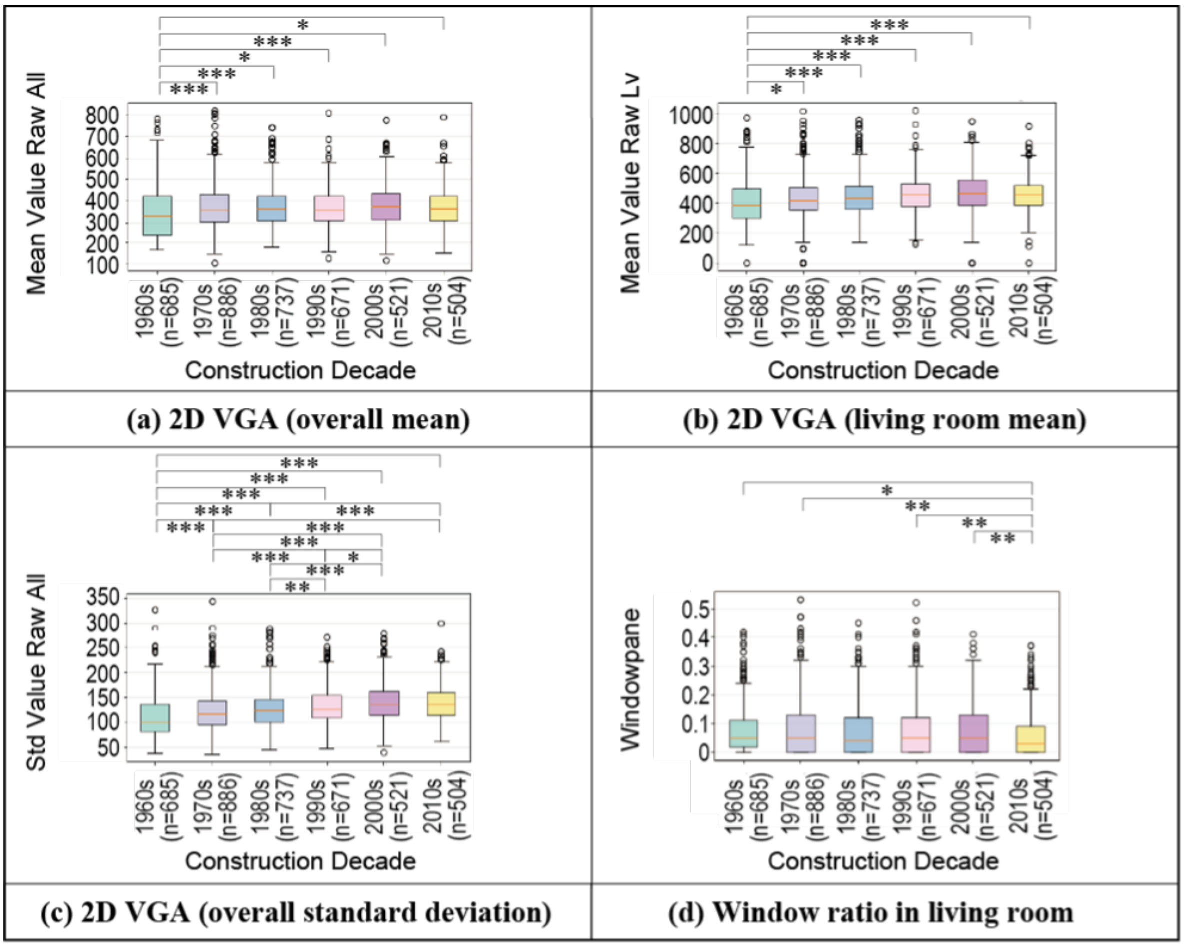


Figure 3.1: Temporal trends of openness indicators by construction decade showing (a) 2D openness scores for entire house, (b) 2D visibility scores for living rooms, (c) Standard deviation of 2D openness, and (d) 3D openness indicators (ceiling and window proportions). (***: 0.1%, **: 1%, *:5%)

Table 3.1 summarizes the one-way ANOVA results for each openness indicator across construction decades. The indicators show whether there are statistically significant differences between decades.

Indicator	**F-statistic**	**P-value**	**Significance**	η^2	**Effect Size**
Overall mean	5.87	0.000021	Yes	0.0073	Very small
Living room mean	17.15	0.000000	Yes	0.0210	Small
Overall std	51.86	0.000000	Yes	0.0609	Medium
Window ratio	3.63	0.002832	Yes	0.0045	Very small

Table 3.1: One-way ANOVA results for openness indicators across construction decades (n=4,004, 6 groups analyzed).

The temporal effect is observed in the overall standard deviation of openness ($\eta^2 = 0.0609$, medium effect), indicating that the variation in spatial layout within properties has changed substantially over time. Living room mean openness shows a small but notable effect ($\eta^2 = 0.0210$), while overall mean openness and window ratio demonstrate very small effects despite statistical significance. To understand the temporal changes of each indicator, the following sections discuss the results of each indicator in more detail.

3.1.1 Pairwise Analysis of Temporal Changes for Each Indicator

We conducted Tukey's HSD post-hoc tests for these indicators. This analysis reveals which specific decades differ significantly from each other and helps identify the direction and magnitude of changes over time.

Mean visibility value of whole property

As shown in Table 3.2, this pattern reveals a clear temporal discontinuity: the 1960s properties demonstrate consistently higher mean visibility values with differences ranging from 4.88% to 6.53% compared to all subsequent decades. The effect sizes are generally small to negligible, but the consistent direction of differences suggests a meaningful architectural shift. After the 1970s, spatial openness characteristics remained roughly stable, with all

Comparison	Mean Diff.	% Diff.	Cohen's d	Effect Size	P-value	Sig.
1960s vs 1970s	-24.39	-6.49%	-0.2139	Small	0.0000	Yes
1960s vs 1980s	-21.33	-5.73%	-0.1930	Negligible	0.0007	Yes
1960s vs 1990s	-18.03	-4.88%	-0.1663	Negligible	0.0107	Yes
1960s vs 2000s	-24.53	-6.53%	-0.2159	Small	0.0003	Yes
1960s vs 2010s	-19.00	-5.13%	-0.1705	Negligible	0.0141	Yes
1970s vs 1980s	+3.06	+0.82%	+0.0313	Negligible	0.9897	No
1970s vs 1990s	+6.36	+1.72%	+0.0670	Negligible	0.8107	No
1970s vs 2000s	-0.14	-0.04%	-0.0014	Negligible	1.0000	No
1970s vs 2010s	+5.39	+1.46%	+0.0560	Negligible	0.9264	No
1980s vs 1990s	+3.30	+0.89%	+0.0374	Negligible	0.9893	No
1980s vs 2000s	-3.20	-0.85%	-0.0349	Negligible	0.9933	No
1980s vs 2010s	+2.33	+0.63%	+0.0262	Negligible	0.9986	No
1990s vs 2000s	-6.50	-1.73%	-0.0741	Negligible	0.8725	No
1990s vs 2010s	-0.97	-0.26%	-0.0115	Negligible	1.0000	No
2000s vs 2010s	+5.53	+1.49%	+0.0626	Negligible	0.9485	No

Table 3.2: Pairwise comparisons (Tukey HSD) for mean visibility value of whole property across construction decades, showing mean differences, effect sizes, and statistical significance.

pairwise differences being non-significant. This suggests that the significant shift in spatial design philosophy occurred between the 1960s and 1970s, likely reflecting the transition from post-war reconstruction architectural practices to more standardized residential design approaches that have persisted through the 2010s.

Mean visibility value of living room

As shown in Table 3.3, the analysis reveals a clear temporal trend in living room openness: properties built in the 1960s consistently demonstrate higher visibility values compared to all subsequent decades, with differences from 4.37% (vs. 1970s) to 11.72% (vs. 2000s). This decline shows a progressive pattern through the 1980s, 1990s, and 2000s, with effect sizes increasing from negligible to small. Notably, the 2010s show no significant difference from the 1990s, suggesting a stabilization of living room design characteristics since the 1990s. The 1970s properties show significantly lower openness compared to later decades (1990s, 2000s, 2010s), while the 1980s only differ significantly from the 2000s. This pattern suggests that

Comparison	Mean Diff.	% Diff.	Cohen's d	Effect Size	P-value	Sig.
1960s vs 1970s	-18.79	-4.37%	-0.1344	Negligible	0.0400	Yes
1960s vs 1980s	-31.80	-7.18%	-0.2350	Small	0.0000	Yes
1960s vs 1990s	-46.57	-10.18%	-0.3495	Small	0.0000	Yes
1960s vs 2000s	-54.53	-11.72%	-0.3888	Small	0.0000	Yes
1960s vs 2010s	-46.48	-10.16%	-0.3477	Small	0.0000	Yes
1970s vs 1980s	-13.01	-2.94%	-0.1035	Negligible	0.3031	No
1970s vs 1990s	-27.79	-6.07%	-0.2252	Small	0.0002	Yes
1970s vs 2000s	-35.74	-7.68%	-0.2775	Small	0.0000	Yes
1970s vs 2010s	-27.70	-6.05%	-0.2261	Small	0.0012	Yes
1980s vs 1990s	-14.77	-3.23%	-0.1270	Negligible	0.2396	No
1980s vs 2000s	-22.73	-4.88%	-0.1865	Negligible	0.0204	Yes
1980s vs 2010s	-14.68	-3.21%	-0.1285	Negligible	0.3337	No
1990s vs 2000s	-7.96	-1.71%	-0.0672	Negligible	0.8890	No
1990s vs 2010s	+0.09	+0.02%	+0.0008	Negligible	1.0000	No
2000s vs 2010s	+8.05	+1.76%	+0.0691	Negligible	0.9108	No

Table 3.3: Pairwise comparisons (Tukey HSD) for mean visibility value of living room across construction decades, showing mean differences, effect sizes, and statistical significance.

living room design became progressively more compartmentalized from the 1960s through the 2000s, potentially reflecting changing privacy preferences, space optimization strategies, or regulatory changes in residential architecture.

Standard deviation of visibility values of whole property

As shown in Table 3.4, the analysis reveals a strong temporal trend in spatial variability: properties built in the 1960s consistently show higher standard deviation of visibility values compared to all subsequent decades, with differences ranging from 7.21% (vs. 1970s) to 19.52% (vs. 2000s). This decline indicates a progressive reduction in spatial diversity within properties, suggesting that modern residential designs feature more functionally specialized rooms with reduced visual accessibility due to more detailed room design. The effect sizes progressively increase from small (1970s-1980s comparisons) to medium (1990s onwards comparisons with 1960s), indicating that this spatial compartmentalization became more pronounced over time. Notably, the 1990s and 2010s show no significant difference, and the 2000s and 2010s are virtually identical, suggesting that spatial compartmentaliza-

Comparison	Mean Diff.	% Diff.	Cohen's d	Effect Size	P-value	Sig.
1960s vs 1970s	-8.70	-7.21%	-0.2204	Small	0.0001	Yes
1960s vs 1980s	-13.77	-10.95%	-0.3556	Small	0.0000	Yes
1960s vs 1990s	-20.88	-15.71%	-0.5382	Medium	0.0000	Yes
1960s vs 2000s	-27.17	-19.52%	-0.6740	Medium	0.0000	Yes
1960s vs 2010s	-26.18	-18.94%	-0.6851	Medium	0.0000	Yes
1970s vs 1980s	-5.07	-4.03%	-0.1392	Negligible	0.0654	No
1970s vs 1990s	-12.17	-9.16%	-0.3347	Small	0.0000	Yes
1970s vs 2000s	-18.47	-13.27%	-0.4924	Small	0.0000	Yes
1970s vs 2010s	-17.47	-12.64%	-0.4917	Small	0.0000	Yes
1980s vs 1990s	-7.11	-5.35%	-0.2019	Small	0.0043	Yes
1980s vs 2000s	-13.40	-9.63%	-0.3686	Small	0.0000	Yes
1980s vs 2010s	-12.41	-8.98%	-0.3644	Small	0.0000	Yes
1990s vs 2000s	-6.30	-4.52%	-0.1734	Negligible	0.0411	Yes
1990s vs 2010s	-5.30	-3.83%	-0.1566	Negligible	0.1444	No
2000s vs 2010s	+1.00	+0.72%	+0.0283	Negligible	0.9981	No

Table 3.4: Pairwise comparisons (Tukey HSD) for standard deviation of visibility values of whole property across construction decades, showing mean differences, effect sizes, and statistical significance.

tion reached a plateau by the 1990s. This pattern likely reflects the evolution toward more compartmentalized floor plans with increased room specialization, where living areas, bedrooms, and other functional spaces became more clearly delineated and visually separated, contrasting with the more spatially integrated and varied arrangements characteristic of 1960s housing.

Window Ratio

As shown in Table 3.5, the post-hoc analysis reveals a clear temporal pattern: the 2010s properties show significantly higher window ratios compared to the 1960s, 1970s, 1990s, and 2000s, with effect sizes ranging from negligible to small. The 2010s demonstrate the highest window proportions, with increases of 26.84% compared to the 1960s, 29.51% compared to the 1970s, 24.12% compared to the 1990s, and 30.54% compared to the 2000s. This increase in window proportions in recent constructions likely reflects modern design preferences emphasizing natural light, improved construction technologies allowing for larger window in-

Comparison	Mean Diff.	% Diff.	Cohen's d	Effect Size	P-value	Sig.
1960s vs 1970s	-0.0017	-2.06%	-0.0189	Negligible	0.9991	No
1960s vs 1980s	+0.0055	+7.49%	+0.0652	Negligible	0.8472	No
1960s vs 1990s	+0.0017	+2.19%	+0.0195	Negligible	0.9993	No
1960s vs 2000s	-0.0023	-2.84%	-0.0273	Negligible	0.9977	No
1960s vs 2010s	+0.0166	+26.84%	+0.2104	Small	0.0155	Yes
1970s vs 1980s	+0.0071	+9.76%	+0.0785	Negligible	0.5757	No
1970s vs 1990s	+0.0033	+4.34%	+0.0359	Negligible	0.9761	No
1970s vs 2000s	-0.0006	-0.79%	-0.0069	Negligible	1.0000	No
1970s vs 2010s	+0.0183	+29.51%	+0.2075	Small	0.0025	Yes
1980s vs 1990s	-0.0038	-4.93%	-0.0422	Negligible	0.9655	No
1980s vs 2000s	-0.0078	-9.61%	-0.0880	Negligible	0.6312	No
1980s vs 2010s	+0.0111	+18.00%	+0.1330	Negligible	0.2362	No
1990s vs 2000s	-0.0040	-4.92%	-0.0437	Negligible	0.9712	No
1990s vs 2010s	+0.0149	+24.12%	+0.1729	Negligible	0.0441	Yes
2000s vs 2010s	+0.0189	+30.54%	+0.2257	Small	0.0072	Yes

Table 3.5: Pairwise comparisons (Tukey HSD) for window ratio across construction decades, showing mean differences, effect sizes, and statistical significance.

stallations, and changing lifestyle preferences that value bright, well-lit living spaces. This trend may also reflect advances in window technology, improved insulation materials, and energy-efficient glass that make larger windows more practical from both comfort and energy efficiency perspectives.

3.2 Geospatial Distribution of Openness Indicators

Figure 3.2 shows the geospatial distribution of representative openness indicators in Tokyo's 23 wards. Of the 4,004 properties analyzed, approximately 3,800 properties with postal code-level address information were selected for geospatial analysis. This study applies ArcGIS Pro 3D Analyst's default Ordinary Kriging: first, an empirical semivariogram is fitted (estimating nugget, range, and sill) to quantify spatial autocorrelation; second, that model is used to compute optimized weights—reflecting both distance and spatial configuration—to interpolate values at unsampled locations and generate a prediction-variance surface, without assuming any explicit spatial trend.

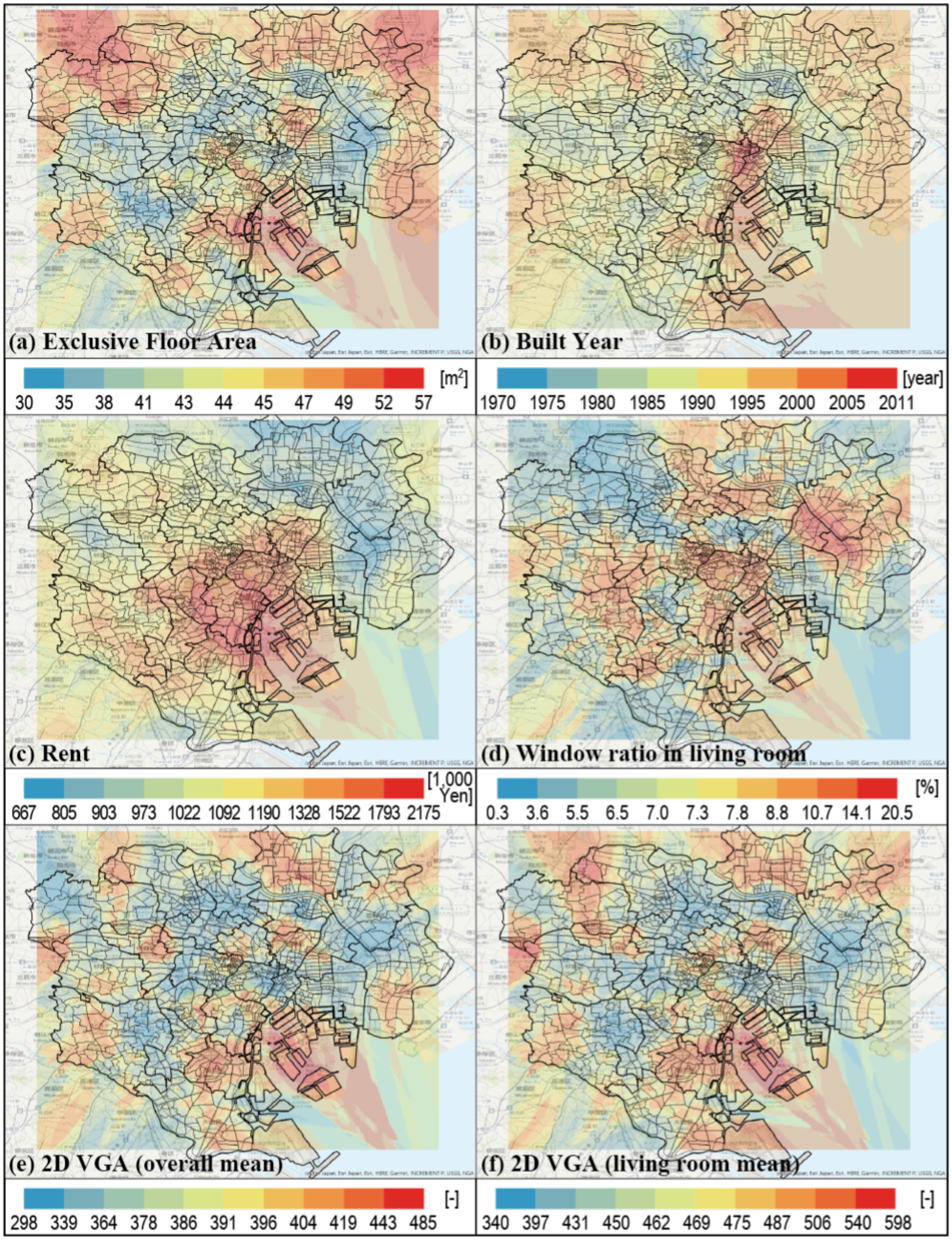


Figure 3.2: Geospatial distribution maps showing: (a) Property rent levels, (b) Construction year distribution, (c) Floor area patterns, (d) Living room window ratios, (e) 2D openness for living rooms, and (f) 2D openness for entire houses.

The spatial patterns of openness represented by overall and living room mean visibility values, rent, construction year, and floor area are partially correlated, indicating that urban redevelopment and housing design trends are manifesting geographically. This correlation suggests that certain areas of Tokyo have experienced similar development and design philosophies.

The ratio of living room windows tends to be high in center areas (6d). This pattern is possible to be influenced by the existence of high-rise apartments where views are valued. In central Tokyo, the premium on natural light and city views drives building constructors to increase the window space, especially in living rooms where residents spend most of their waking hours. The emphasis on windows in these areas also reflects the higher property values that can justify the additional costs of better views.

Areas with high visibility values are scattered throughout the southern part of the city center and other areas (6e and 6f). This scattered distribution suggests that open floor plans are not concentrated in any single district but rather reflect various factors including development timing, target demographics, and local planning regulations. The southern areas may benefit from better sunlight exposure, making open layouts more attractive and practical.

The relatively low visibility values in suburban areas are thought to be influenced by housing types with many individual rooms and housing complexes. Suburban housing often reflects traditional Japanese housing preferences that prioritize privacy and clearly defined spaces for different activities. These areas may also have more families with children, where separated rooms serve practical purposes for study, sleep, and family activities.

3.3 Correlation Analysis Among Indicators and with Rent and Others

Figure 3.3 shows the correlation between 2D/3D openness indicators, between openness indicators and rent, along with other indicators. This analysis helps us understand how different aspects of spatial openness relate to each other and to market factors.

No clear correlation was found between the 2D openness indicator and the 3D openness indicator, namely the window ratio VS. the mean visibility value of the properties (7a). This finding suggests that floor plan openness and vertical spatial characteristics operate independently in residential design. A house might have an open floor plan but low window ratio, or conversely, have separated rooms but high ceilings and large windows. This independence indicates that designers and residents can achieve different types of openness through various design strategies.

On the other hand, a high correlation was confirmed between the mean value and standard deviation of 2D openness (7b). This relationship indicates that properties with higher average 2D openness also tend to have greater variation in openness across different spaces. This pattern makes sense from a design perspective, as houses with very open living areas often offers larger area and maintain more functionality focused, enclosed spaces for perhaps multiple bedrooms and bathrooms, creating a higher deivations in visibility values between small rooms and large rooms.

The 3D openness indicators were generally complementary to each other (7c). This complementary relationship suggests that different vertical spatial elements work together to create the overall sense of openness in interior spaces. For example, rooms with smaller windows might compensate with higher ceilings or more visible ceiling area to maintain a sense of spaciousness.

Additionally, there was a tendency for properties with higher openness to have higher rent price (7e). This positive correlation indicates that spatial openness is valued in the

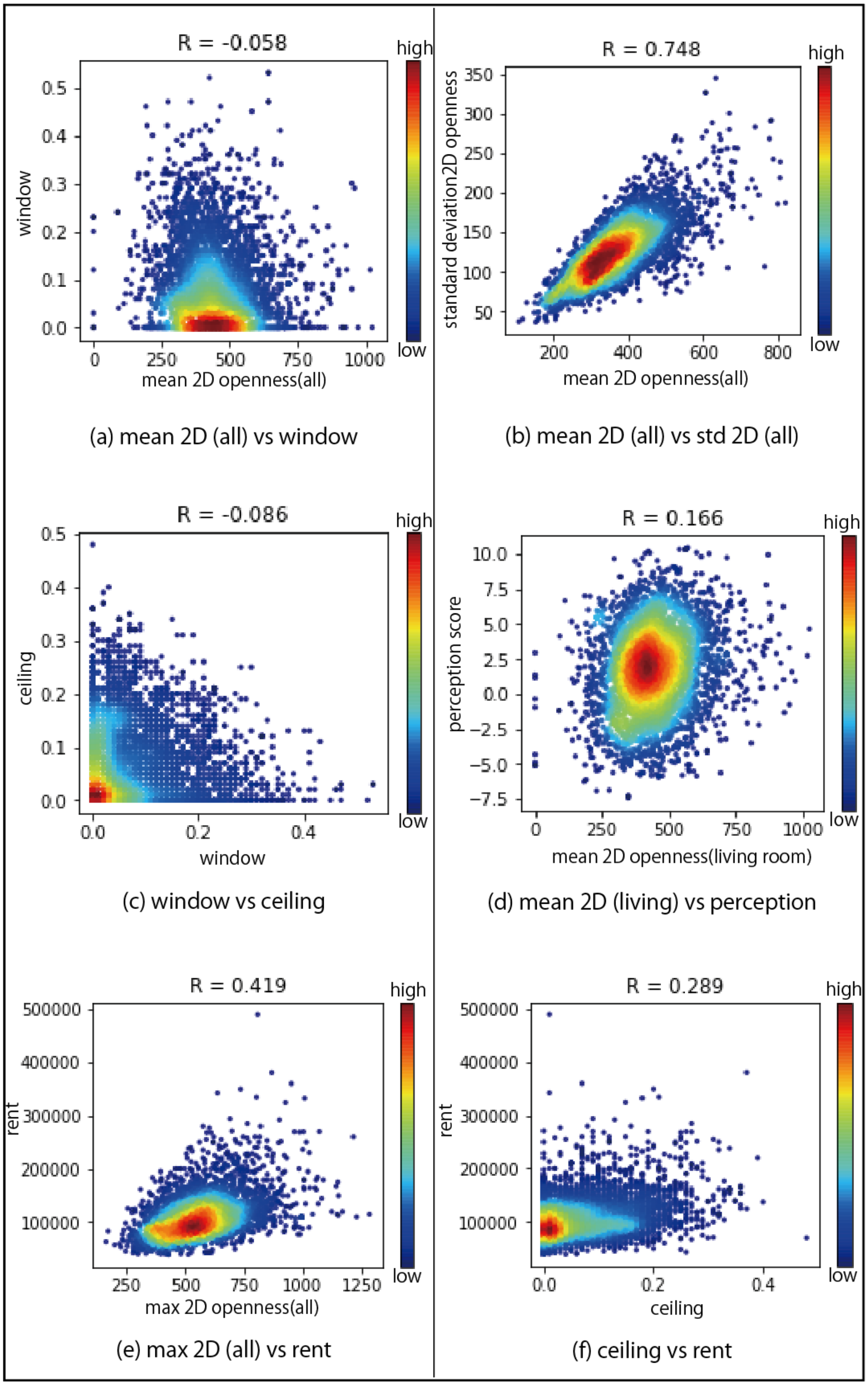


Figure 3.3: Correlation analysis showing: (a) 2D vs 3D openness indicators, (b) Mean vs standard deviation of 2D openness, (c) Relationships among 3D openness indicators, (d) 2D openness vs impression evaluation scores, and (e) Openness indicators vs rent levels.

rental market and commands a premium. Tenants appear willing to pay more for properties that offer better visual connectivity, more natural light, and a greater sense of space. This market preference validates the importance of openness as a design consideration for rental properties.

Here in (d), we introduce a impression evaluation score that presents the user perception goodness from one of the previous research in our lab, which utilizes a trained feature extraction network to extract 1D feature vector from interior images, then input to a preference scoring model that was trained over a set of user annotated dataset. The detailed information can refer to this report [11]. With these impression evaluation scores calculated for all 4,004 properties, we can also investigate the correlation between these scores and 2D openness indicators for living rooms. Comparing these scores with 2D openness indicators for living rooms, there was almost no correlation between them (7d), showing the clear perception gap between a 2D floorplan image and the actual living room photograph to users' eyes.

While spatial openness is one of the elements that characterize a space, this result suggests that other elements such as interior design and furniture may have a greater influence on impressions. The lack of correlation between objective openness measures and subjective impressions reveals the complexity of how people perceive and evaluate living spaces. Factors such as color schemes, lighting quality, furniture style and arrangement, cleanliness, and overall aesthetic appeal may play more significant roles in forming positive impressions than the geometric properties of the space itself.

3.4 Impact of Openness Indicators on Rental Prices

To further investigate the relationship between openness indicators and market value, we developed multivariate linear regression models to fit the rental prices using the various openness indicators. This analysis aims to quantify the economic impact of spatial openness characteristics and identify which specific indicators contribute most significantly to rental price determination.

Fitting with Unselected Features

The predictive modeling process involved selecting relevant features from our comprehensive set of openness indicators. Table 3.6 presents the features for the regression model, categorized by their measurement type and spatial scope.

Category	Feature	Description
Whole Property 2D	min_value_raw_all	Minimum visibility value across all spaces
	max_value_raw_all	Maximum visibility value across all spaces
	mean_value_raw_all	Average visibility value across all spaces
	std_value_raw_all	Standard deviation of visibility values
Living Room 2D	mean_value_raw_lv	Average visibility value in living room
	min_value_raw_lv	Minimum visibility value in living room
	max_value_raw_lv	Maximum visibility value in living room
	std_value_raw_lv	Standard deviation of visibility in living room
3D Openness	windowpane	Proportion of window area
	floor	Proportion of visible floor area
	ceiling	Proportion of visible ceiling area
	wall	Proportion of visible wall area

Table 3.6: Selected features for rental price prediction model, organized by measurement category and spatial scope.

After performing the initial regression fit using these features, we obtained the regression results shown in Figure 3.4. The model achieved an R^2 of 0.3, indicating moderate predictive power in explaining rental price variance through our openness indicators. Residual analysis revealed no clear imbalance in the error distribution, suggesting negligible bias in the linear model fit. The results illustrate the predictive performance and feature importance of our openness indicators for rental price prediction.

Correlation and VIF Checks to Reduce Multicollinearity

The correlations within the features is displayed in Figure 3.5, and the feature importance is displayed in Figure 3.6.

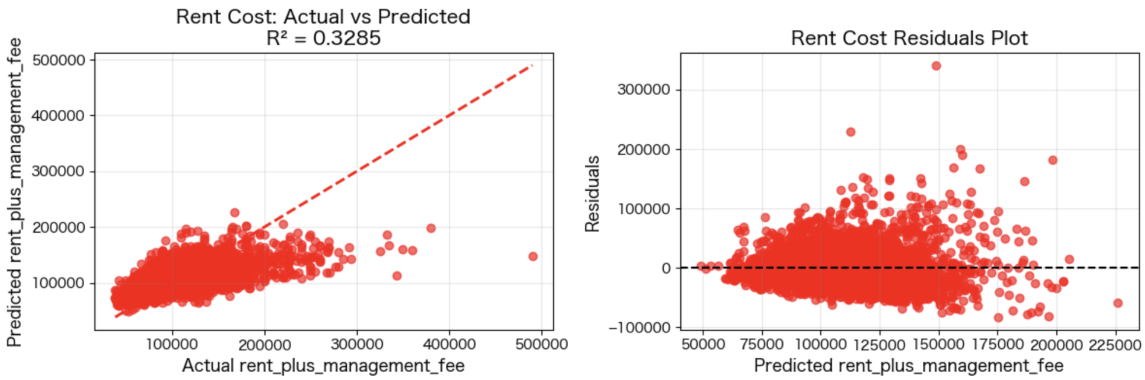


Figure 3.4: Predictive performance and feature importance analysis for rental price prediction using openness indicators. The figure shows R^2 scores and relative importance rankings of different spatial openness measures in determining rental prices.

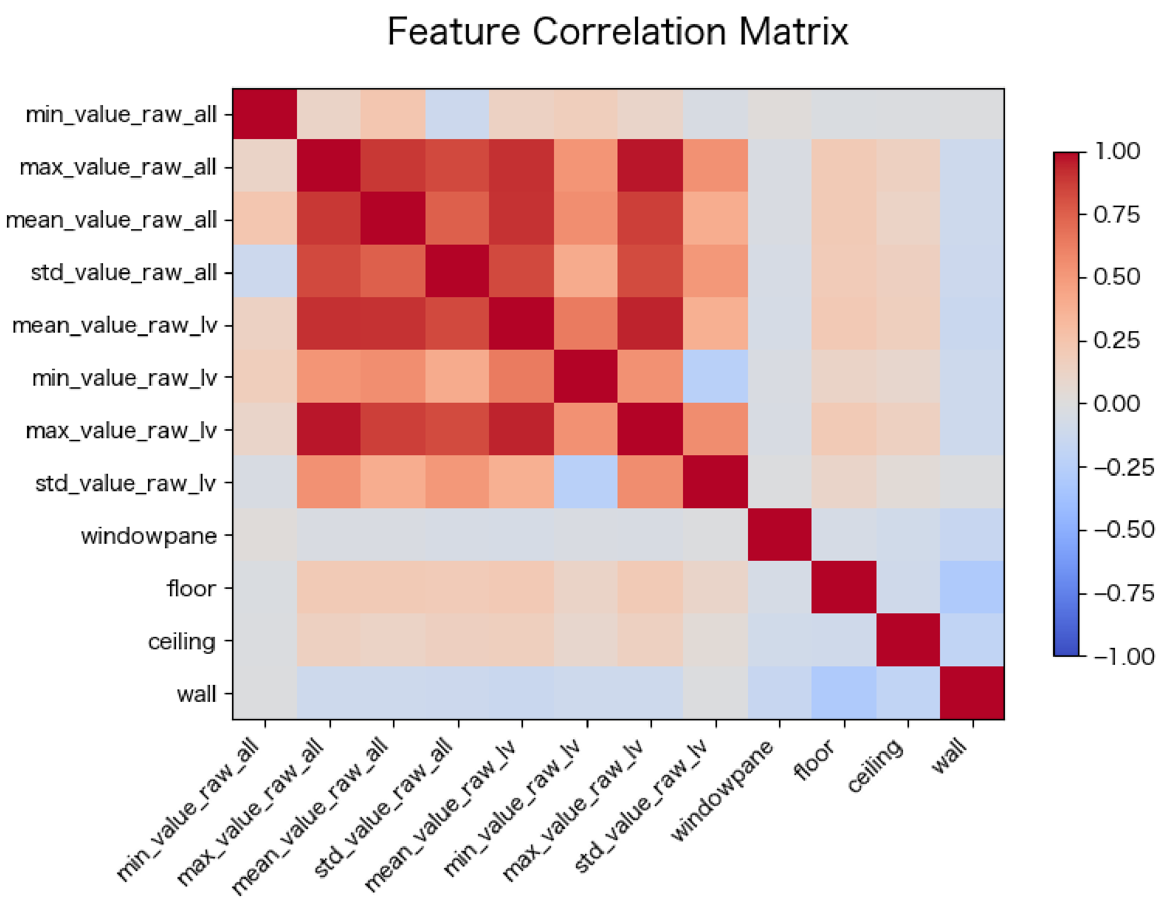


Figure 3.5: Predictive performance of openness indicators for rental price prediction showing R^2 scores and feature importance rankings. The model demonstrates the relative contribution of different spatial openness measures to rental price determination.

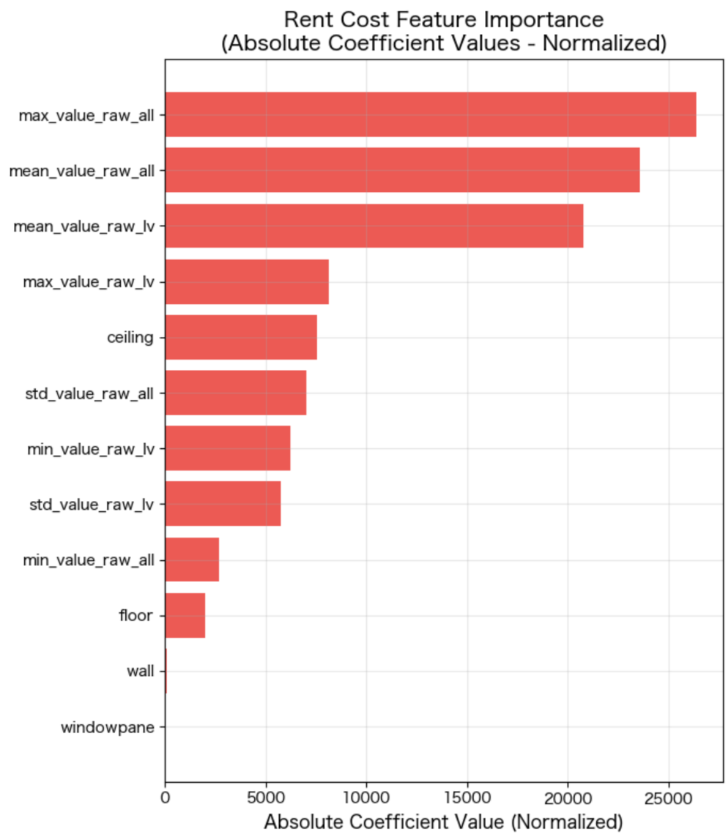


Figure 3.6: Correlation matrix and feature importance analysis for the 12 selected openness indicators used in rental price prediction. The heatmap shows inter-feature correlations while the importance scores indicate each feature's contribution to the predictive model.

To address potential multicollinearity issues observed in the feature correlation analysis, we conducted Variance Inflation Factor (VIF) checks and employed stepwise feature selection to identify the optimal feature subset while maintaining model performance. High correlation between certain features suggested redundancy that could be eliminated without significant loss in predictive power.

The VIF analysis revealed several features with high multicollinearity ($VIF > 5$), particularly among the 2D openness measures within the same spatial scope, as shown in Table 3.7.

Feature	VIF	Status
min_value_raw_all	1.35	Low
max_value_raw_all	28.16	High
mean_value_raw_all	8.34	Moderate
std_value_raw_all	5.58	Moderate
min_value_raw_lv	3.40	Low
max_value_raw_lv	37.94	High
std_value_raw_lv	3.63	Low
mean_value_raw_lv	18.29	High
windowpane	1.06	Low
floor	1.21	Low
ceiling	1.13	Low
wall	1.23	Low

Table 3.7: Variance Inflation Factor (VIF) for multicollinearity assessment, $VIF > 10$: High, $VIF > 5$: Moderate, $VIF < 5$: Low

Stepwise Feature Selection for Optimal Model

Based on the VIF analysis results, we selected the 7 features highlighted in red in Table 3.7 with acceptable multicollinearity levels ($VIF < 10$) for the final model. These selected features include: min_value_raw_all, max_value_raw_all, mean_value_raw_all, std_value_raw_all, std_value_raw_lv, and ceiling. Notably, all the visibility-based openness indicators from whole properties (min_value_raw_all, max_value_raw_all, mean_value_raw_all, std_value_raw_all) were retained after the selection, demonstrating their independence from each other and their distinct contributions to model performance. This refined feature set eliminated the most problematic variables while maintaining strong predictive performance. The results of this 7-feature model are shown in Figure 3.7, demonstrating that the reduced model maintains comparable performance. The updated feature importance is shown in Figure 3.8.

Interpretation of Feature Importance

The overall visibility values of properties demonstrate high importance because they capture the complete spatial connectivity and accessibility patterns which directly relates to the perceived spaciousness and flow that tenants value. In contrast, the living room-specific features show relatively low importance, perhaps due to their partial representation of the property's overall spatial quality. For 3D openness indicators, only ceiling shows relatively good importance, possibly due to its strong correlation with perceived vertical spaciousness. However, we must understand that rent price is a very complicated target to predict and is certainly not impacted solely by our selected features. Therefore, our observations can only serve as potential insights to link the relationship between price and our openness indicators, not aiming to comprehensively predict the rent price under any assumption.

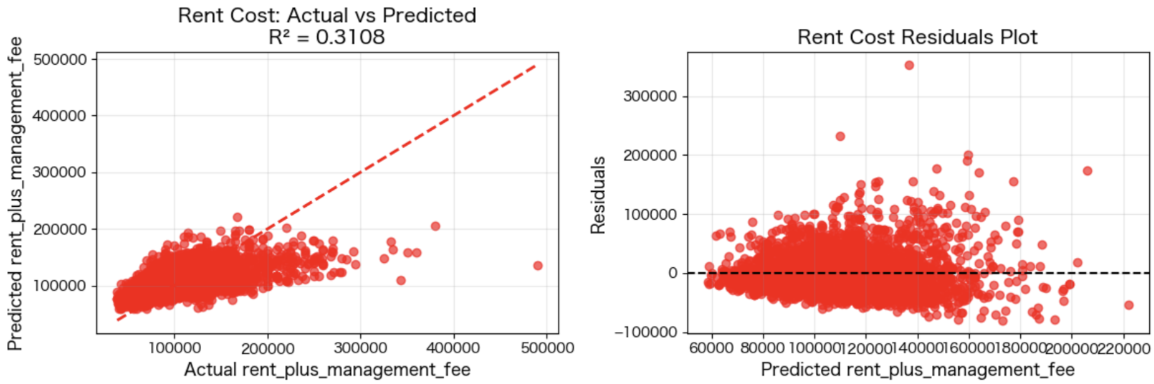


Figure 3.7: Predictive performance of with reduced features

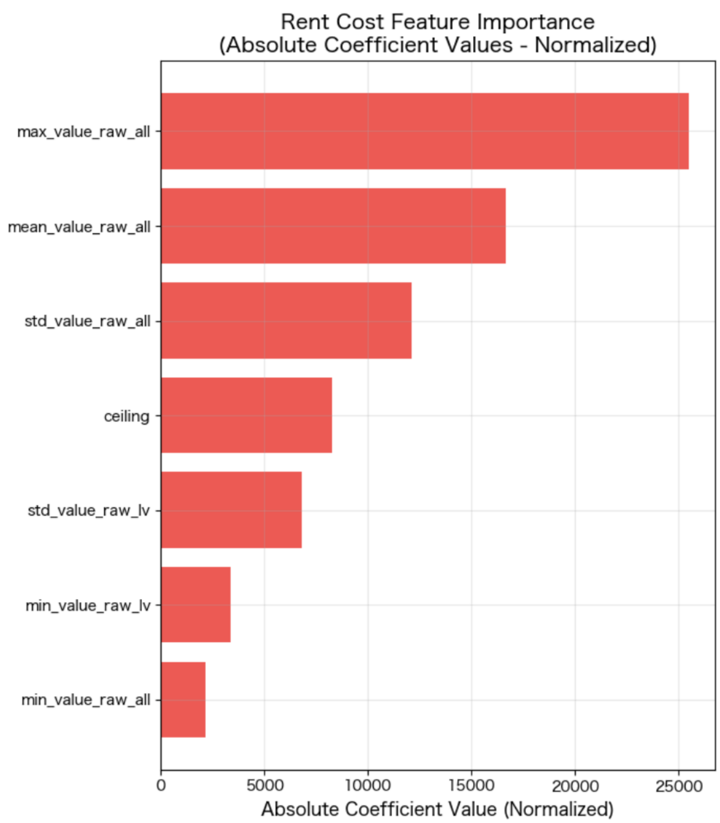


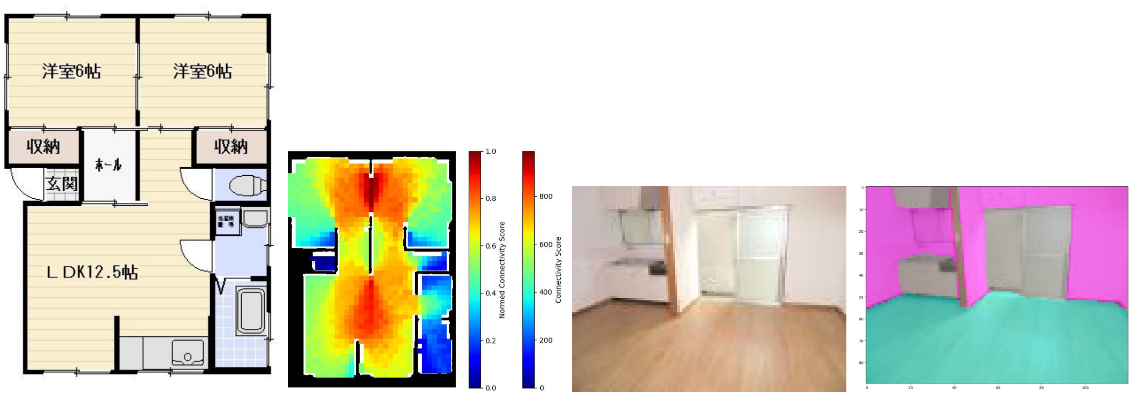
Figure 3.8: Feature importance of the 7 selected features

3.5 Visual Validation of Openness Indicators

To validate the reliability and interpretability of our openness indicators, we present a visual comparison of properties with varying openness characteristics. For each indicator, we collected properties from the top 25% and bottom 25% of the distribution, then sampled 2 properties from each group to showcase the intuitive impressions associated with different indicator values. This section simply showcases floor plans and corresponding living room images. The visual examples demonstrate how the quantitative measures correspond to observable spatial characteristics, providing intuitive understanding of what the numerical indicators represent in actual residential spaces. This validation helps bridge the gap between computational analysis and human perception of spatial openness as an extra validation of the effectiveness of our methodolgies.

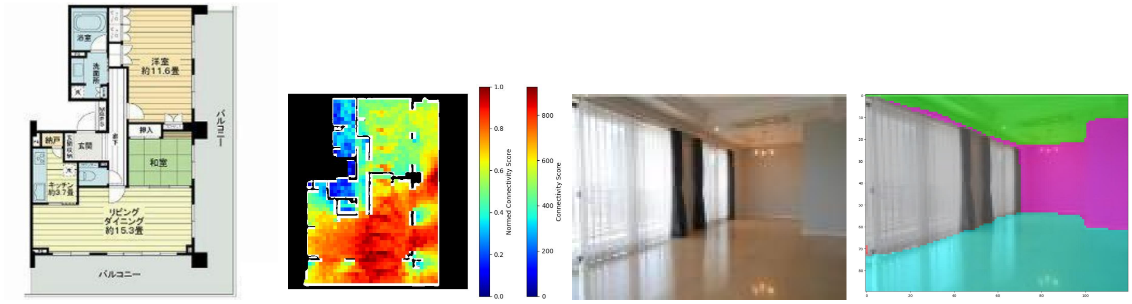
3.5.1 2D Openness Indicators

Mean visibility values - Whole Property



High visibility - Floorplan 1 | High visibility - Heatmap 1 | High visibility - Living Room 1 | High visibility - Segmentation 1

Figure 3.9: High mean visibility property 1 - floor plan, heatmap, living room, and segmentation.



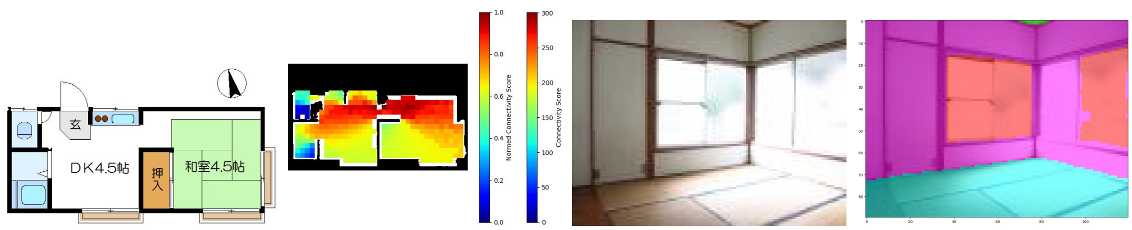
High visibility - Floorplan 2

High visibility - Heatmap 2

High visibility - Living Room 2

High visibility - Segmentation 2

Figure 3.10: High mean visibility property 2 - floor plan, heatmap, living room, and segmentation.



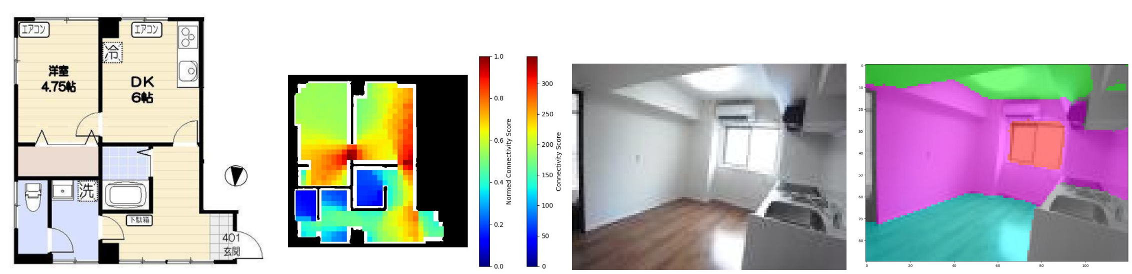
Low visibility - Floorplan 1

Low visibility - Heatmap 1

Low visibility - Living Room 1

Low visibility - Segmentation 1

Figure 3.11: Low mean visibility property 1 - floor plan, heatmap, living room, and segmentation.



Low visibility - Floorplan 2

Low visibility - Heatmap 2

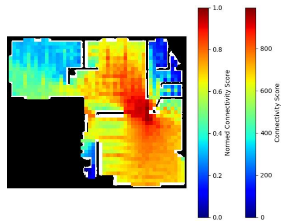
Low visibility - Living Room 2

Low visibility - Segmentation 2

Figure 3.12: Low mean visibility property 2 - floor plan, heatmap, living room, and segmentation.

Standard Deviation of visibility values

High Std - Floorplan 1



High Std - Heatmap 1


High Std - Living Room 1

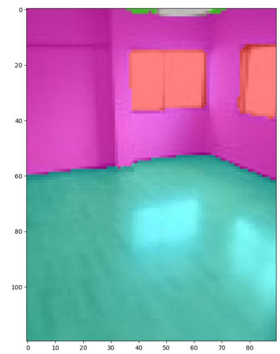
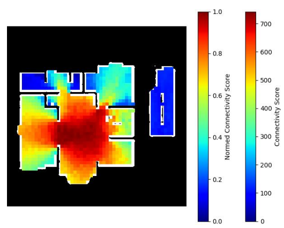
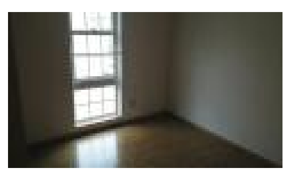

High Std - Segmentation 1

Figure 3.13: High standard deviation property 1 - floor plan, heatmap, living room, and segmentation.

High Std - Floorplan 2



High Std - Heatmap 2


High Std - Living Room 2

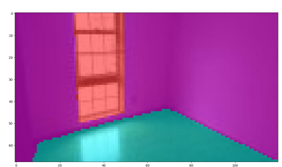
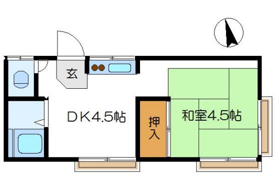
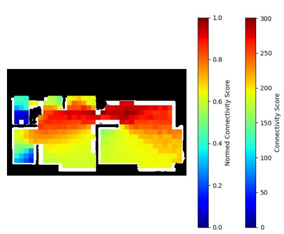

High Std - Segmentation 2

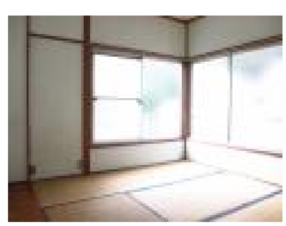
Figure 3.14: High standard deviation property 2 - floor plan, heatmap, living room, and segmentation.



Low Std - Floorplan 1



Low Std - Heatmap 1


Low Std - Living Room 1

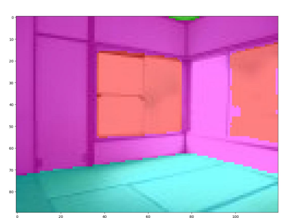
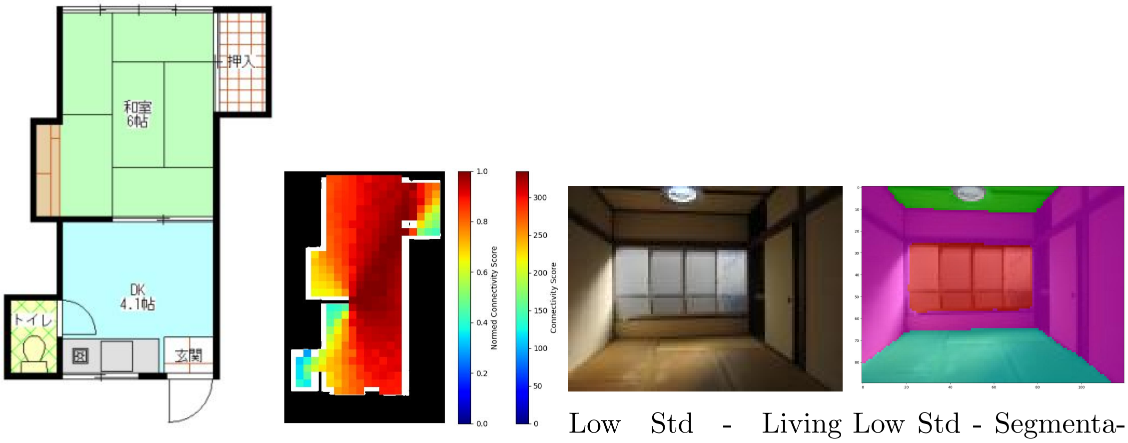

Low Std - Segmentation 1

Figure 3.15: Low standard deviation property 1 - floor plan, heatmap, living room, and segmentation.

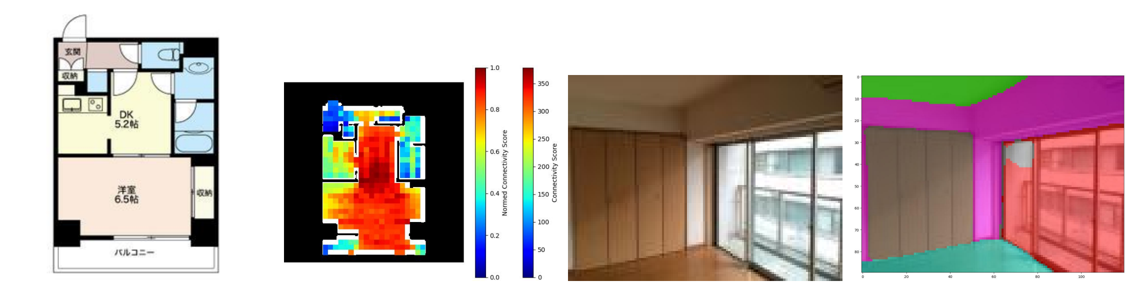


Low Std - Floorplan 2 Low Std - Heatmap 2 Low Std - Living Room 2 Low Std - Segmentation 2

Figure 3.16: Low standard deviation property 2 - floor plan, heatmap, living room, and segmentation.

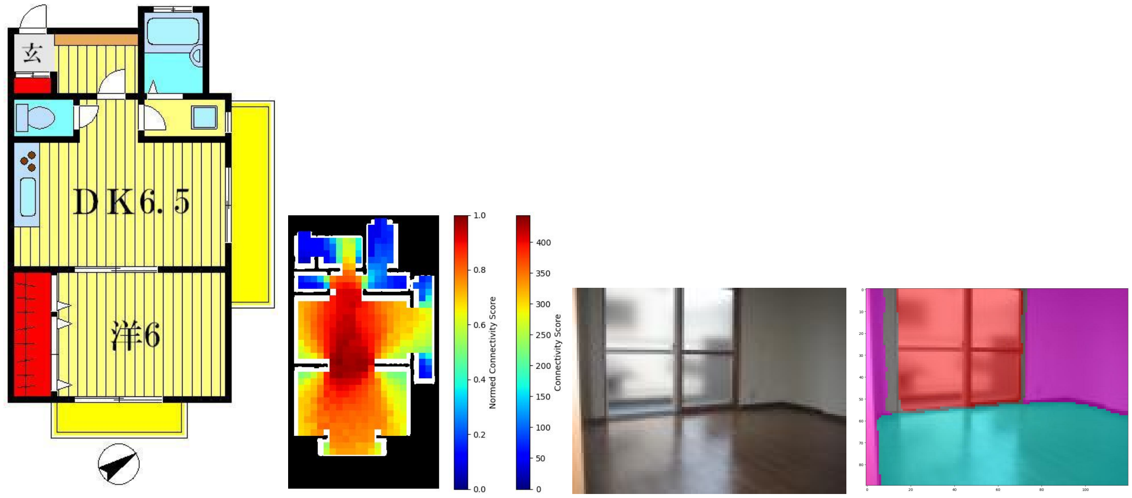
3.5.2 3D Openness Indicators

Window Ratio



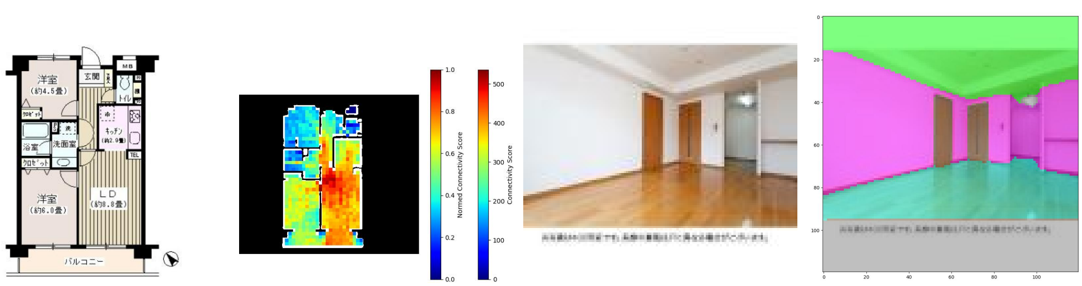
High Window - Floorplan 1 High Window - Heatmap 1 High Window - Living Room 1 High Window - Segmentation 1

Figure 3.17: High window ratio property 1 - floor plan, heatmap, living room, and segmentation.



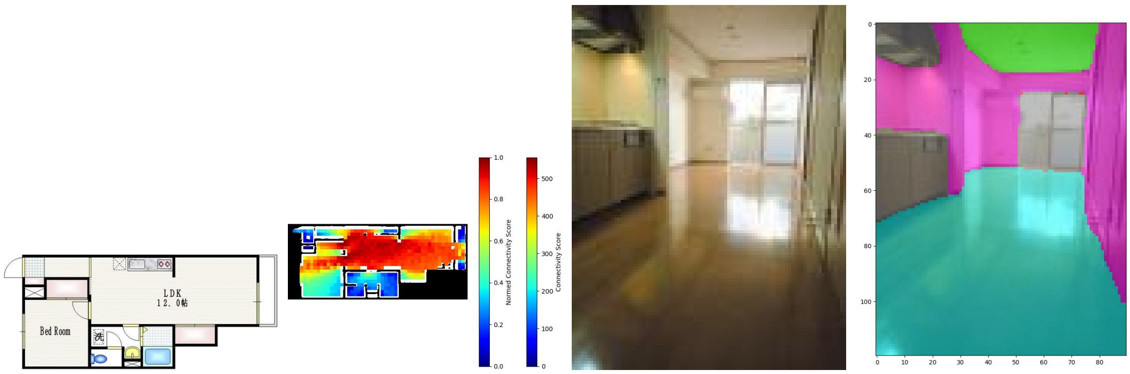
High Window - Floorplan 2 | High Window - Heatmap 2 | High Window - Living Room 2 | High Window - Segmentation 2

Figure 3.18: High window ratio property 2 - floor plan, heatmap, living room, and segmentation.



Low Window - Floorplan 1 | Low Window - Heatmap 1 | Low Window - Living Room 1 | Low Window - Segmentation 1

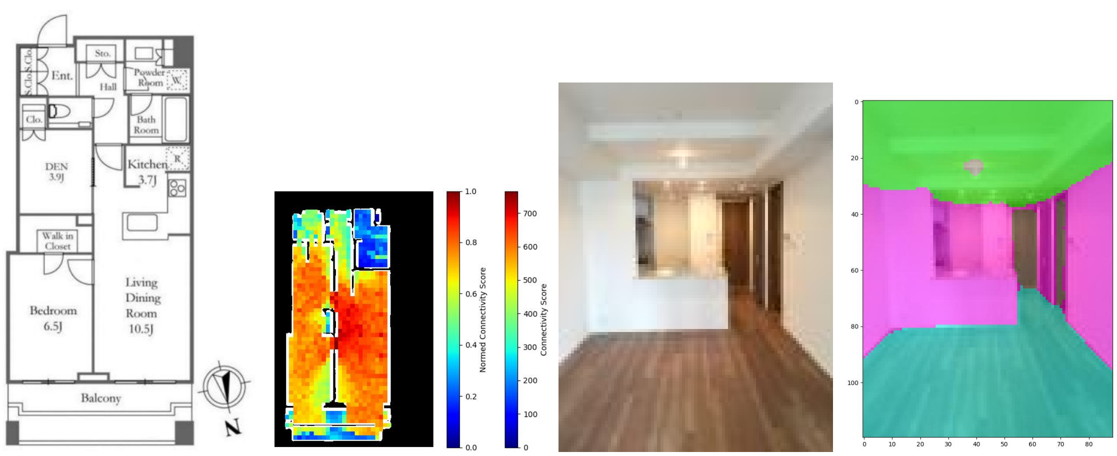
Figure 3.19: Low window ratio property 1 - floor plan, heatmap, living room, and segmentation.



Low Window - Floorplan 2 | Low Window - Heatmap 2 | Low Window - Living Room 2 | Low Window - Segmentation 2

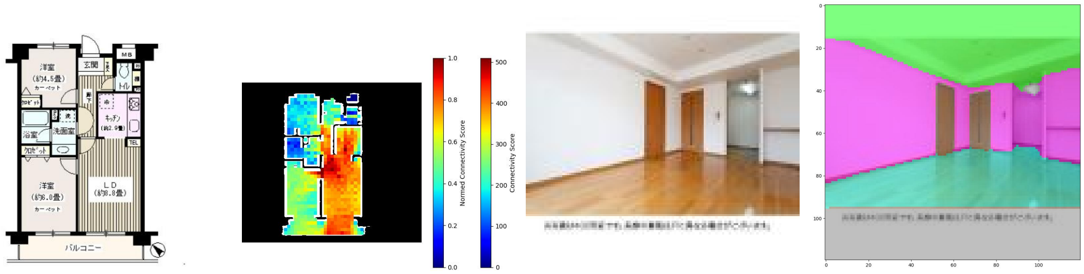
Figure 3.20: Low window ratio property 2 - floor plan, heatmap, living room, and segmentation.

Ceiling Visibility



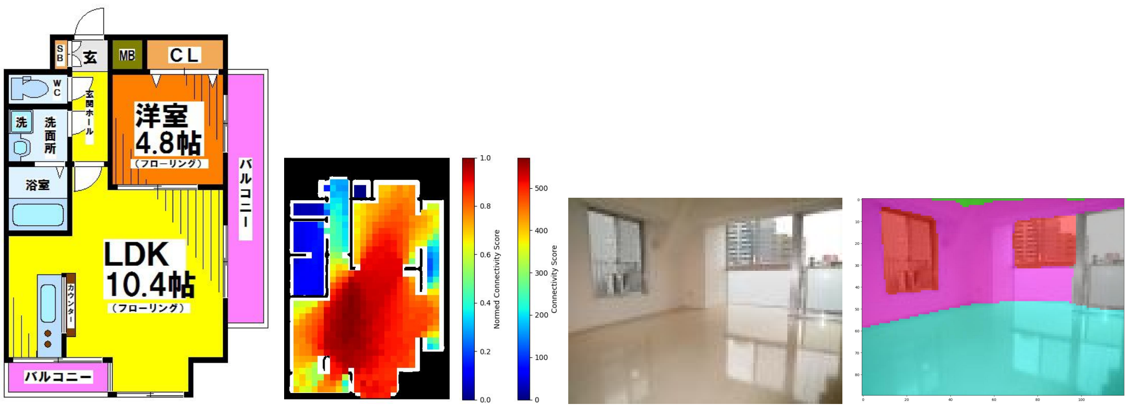
High Ceiling - Floorplan 1 | High Ceiling - Heatmap 1 | High Ceiling - Living Room 1 | High Ceiling - Segmentation 1

Figure 3.21: High ceiling visibility property 1 - floor plan, heatmap, living room, and segmentation.



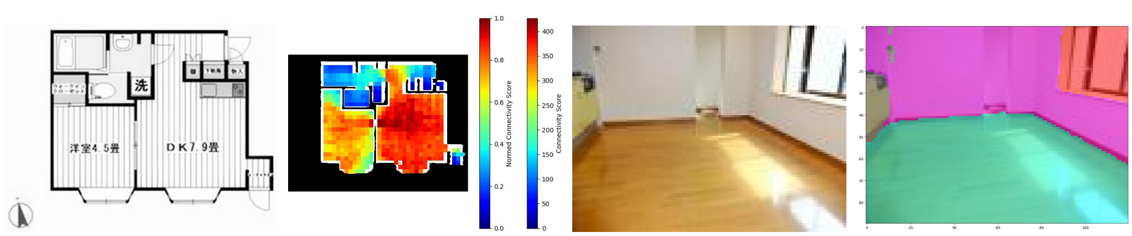
High Ceiling - Floorplan 2 | High Ceiling - Heatmap 2 | High Ceiling - Living Room 2 | High Ceiling - Segmentation 2

Figure 3.22: High ceiling visibility property 2 - floor plan, heatmap, living room, and segmentation.



Low Ceiling - Floorplan 1 | Low Ceiling - Heatmap 1 | Low Ceiling - Living Room 1 | Low Ceiling - Segmentation 1

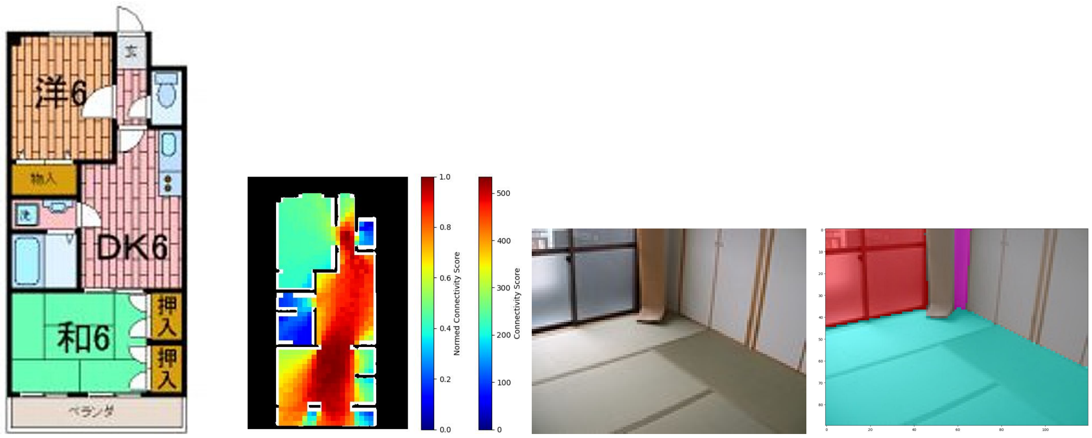
Figure 3.23: Low ceiling visibility property 1 - floor plan, heatmap, living room, and segmentation.



Low Ceiling - Floorplan 2 | Low Ceiling - Heatmap 2 | Low Ceiling - Living Room 2 | Low Ceiling - Segmentation 2

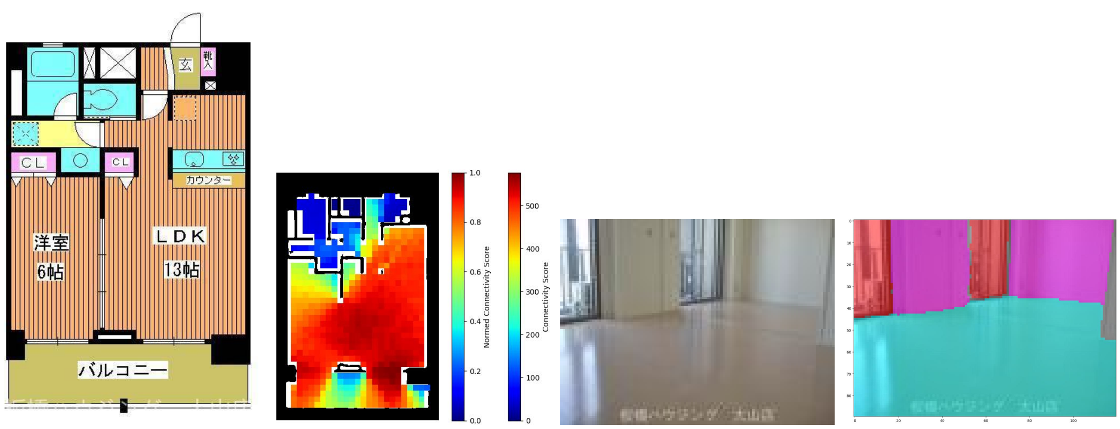
Figure 3.24: Low ceiling visibility property 2 - floor plan, heatmap, living room, and segmentation.

Floor Visibility



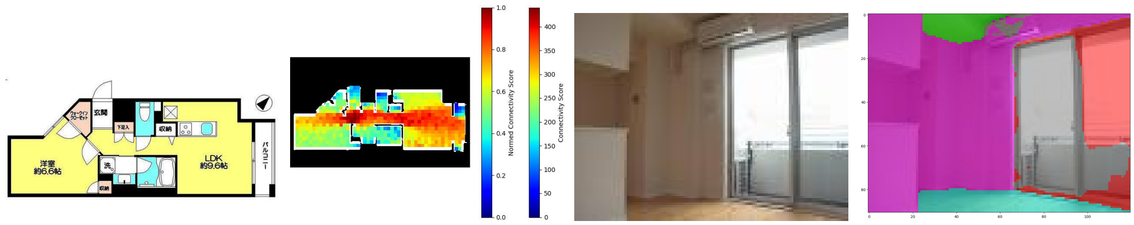
High Floor - Floor-plan 1 | High Floor - Heatmap 1 | High Floor - Living Room 1 | High Floor - Segmentation 1

Figure 3.25: High floor visibility property 1 - floor plan, heatmap, living room, and segmentation.



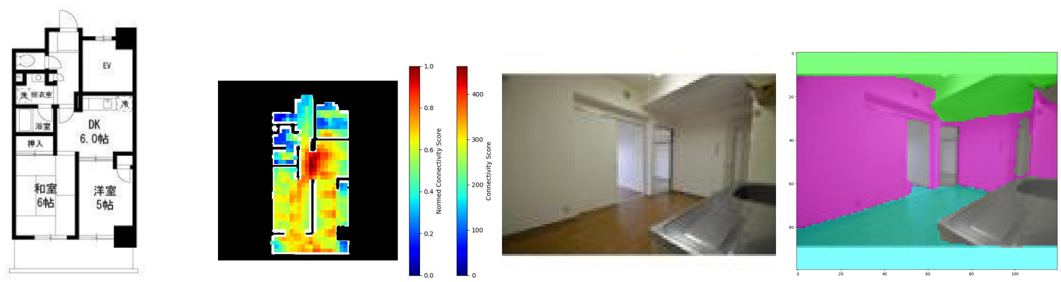
High Floor - Floor-plan 2 | High Floor - Heatmap 2 | High Floor - Living Room 2 | High Floor - Segmentation 2

Figure 3.26: High floor visibility property 2 - floor plan, heatmap, living room, and segmentation.



Low Floor - Floorplan 1 | Low Floor - Heatmap 1 | Low Floor - Living Room 1 | Low Floor - Segmentation 1

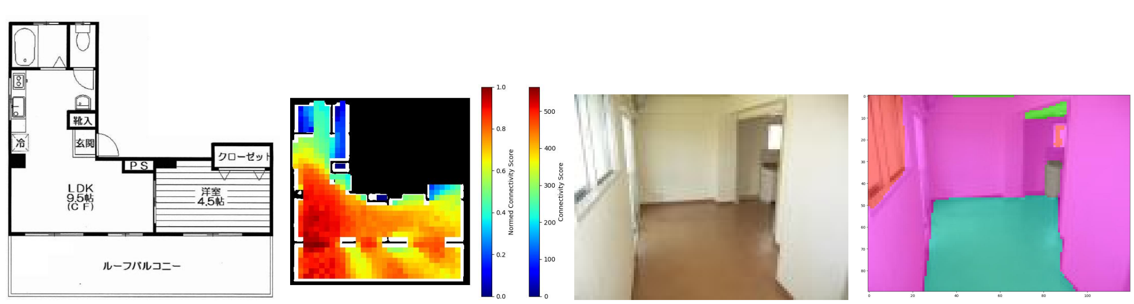
Figure 3.27: Low floor visibility property 1 - floor plan, heatmap, living room, and segmentation.



Low Floor - Floorplan 2 | Low Floor - Heatmap 2 | Low Floor - Living Room 2 | Low Floor - Segmentation 2

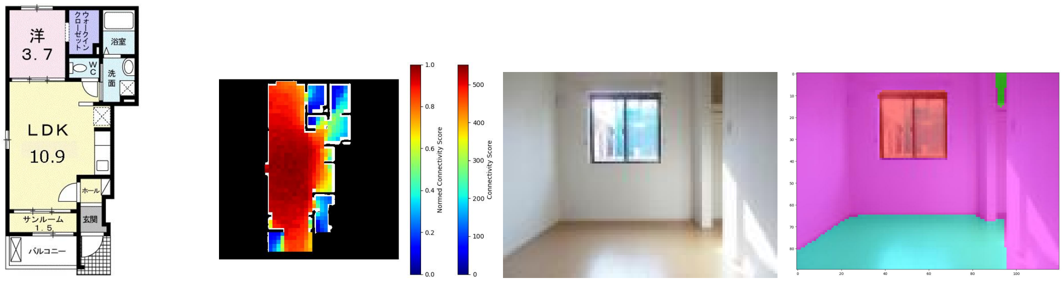
Figure 3.28: Low floor visibility property 2 - floor plan, heatmap, living room, and segmentation.

Wall Visibility



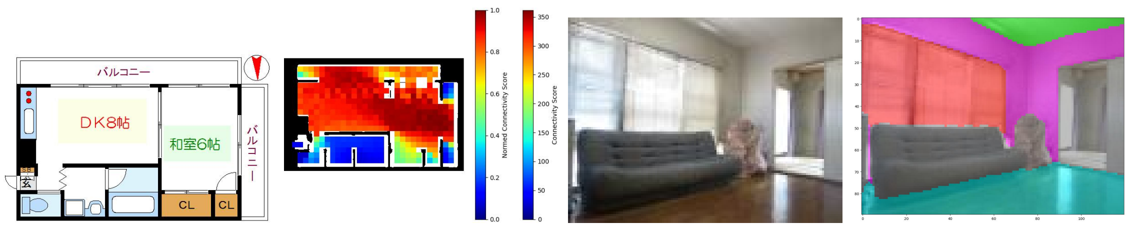
High Wall - Floorplan 1 | High Wall - Heatmap 1 | High Wall - Living Room 1 | High Wall - Segmentation 1

Figure 3.29: High wall visibility property 1 - floor plan, heatmap, living room, and segmentation.



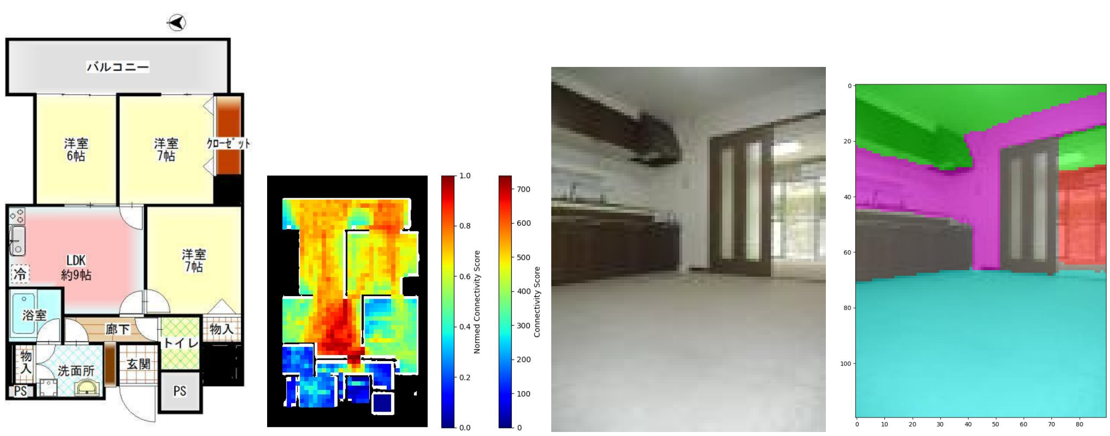
High Wall - Floorplan 2 | High Wall - Heatmap 2 | High Wall - Living Room 2 | High Wall - Segmentation 2

Figure 3.30: High wall visibility property 2 - floor plan, heatmap, living room, and segmentation.



Low Wall - Floorplan 1 | Low Wall - Heatmap 1 | Low Wall - Living Room 1 | Low Wall - Segmentation 1

Figure 3.31: Low wall visibility property 1 - floor plan, heatmap, living room, and segmentation.

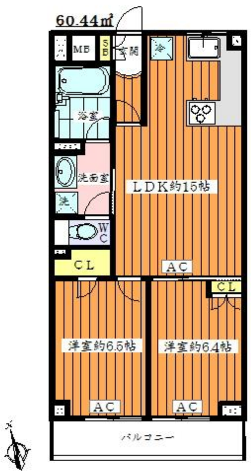


Low Wall - Floorplan 2 | Low Wall - Heatmap 2 | Low Wall - Living Room 2 | Low Wall - Segmentation 2

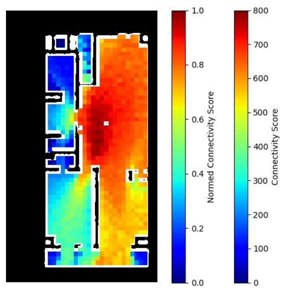
Figure 3.32: Low wall visibility property 2 - floor plan, heatmap, living room, and segmentation.

3.5.3 Combined Indicator Analysis

High visibility values with Low Window Ratio



High visibility Low Window - Floorplan 1



High visibility Low Window - Heatmap 1


High visibility Low Window - Interior 1

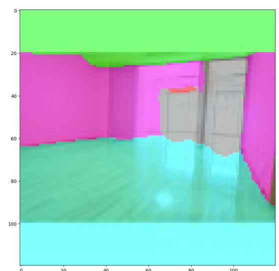
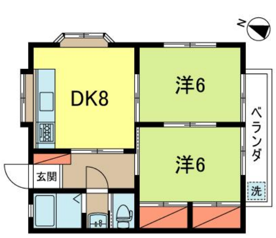
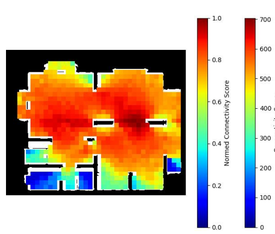
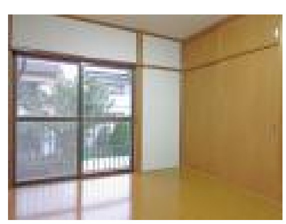

High visibility Low Window - Segmentation 1

Figure 3.33: High visibility low window property 1 - floor plan, heatmap, interior, and segmentation.



High visibility Low Window - Floorplan 2


High visibility Low Window - Heatmap 2

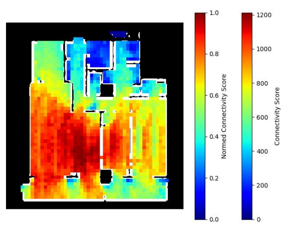

High visibility Low Window - Interior 2


High visibility Low Window - Segmentation 2

Figure 3.34: High visibility low window property 2 - floor plan, heatmap, interior, and segmentation.

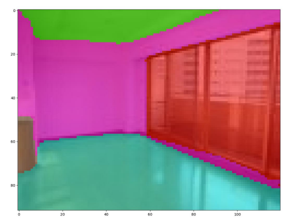
High visibility values with High Window Ratio

High visibility High Window - Floorplan 1



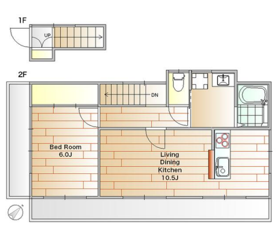
High visibility High Window - Heatmap 1

High visibility High Window - Interior 1

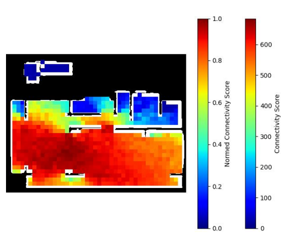


High visibility High Window - Segmentation 1

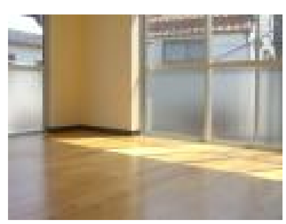
Figure 3.35: High visibility high window property 1 - floor plan, heatmap, interior, and segmentation.



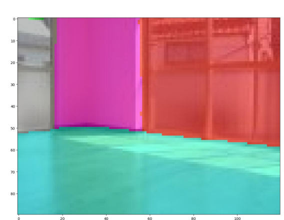
High visibility High Window - Floorplan 2



High visibility High Window - Heatmap 2



High visibility High Window - Interior 2

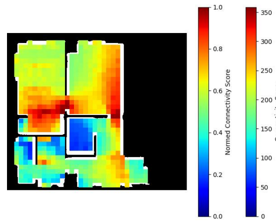


High visibility High Window - Segmentation 2

Figure 3.36: High visibility high window property 2 - floor plan, heatmap, interior, and segmentation.

Low visibility values with High Window Ratio

Low visibility High Window - Floorplan 1



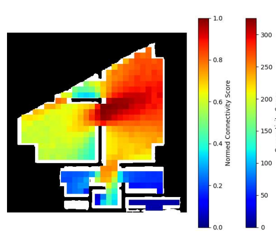
Low visibility High Window - Heatmap 1

Low visibility High Window - Interior 1

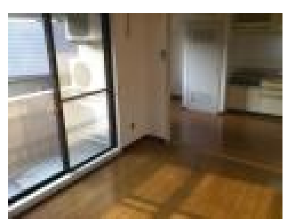
Low visibility High Window - Segmentation 1

Figure 3.37: Low visibility high window property 1 - floor plan, heatmap, interior, and segmentation.

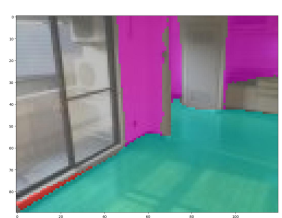
Low visibility High Window - Floorplan 2



Low visibility High Window - Heatmap 2



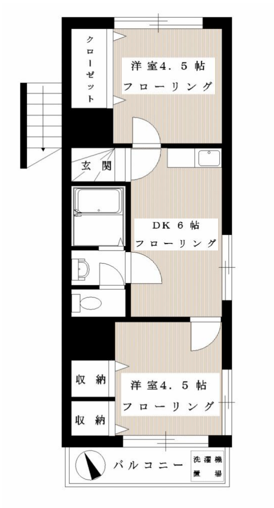
Low visibility High Window - Interior 2



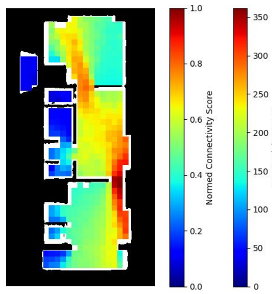
Low visibility High Window - Segmentation 2

Figure 3.38: Low visibility high window property 2 - floor plan, heatmap, interior, and segmentation.

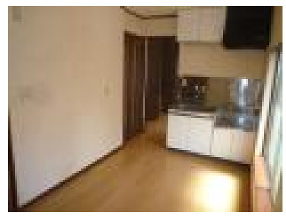
Low visibility values with Low Window Ratio



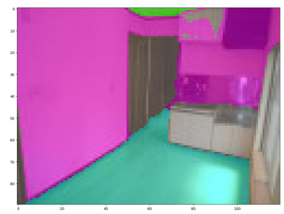
Low visibility Low Window - Floorplan 1



Low visibility Low Window - Heatmap 1



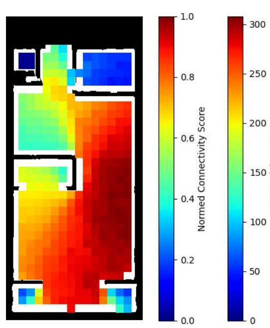
Low visibility Low Window - Interior 1



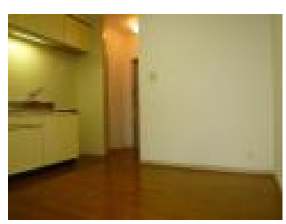
Low visibility Low Window - Segmentation 1

Figure 3.39: Low visibility low window property 1 - floor plan, heatmap, interior, and segmentation.

Low visibility Low Window - Floorplan 2



Low visibility Low Window - Heatmap 2



Low visibility Low Window - Interior 2

Low visibility Low Window - Segmentation 2

Figure 3.40: Low visibility low window property 2 - floor plan, heatmap, interior, and segmentation.

Chapter 4

Conclusions

4.1 Summary of Key Findings

This thesis presented a comprehensive methodology for quantifying spatial openness in residential properties through the integration of computer vision techniques and large-scale data analysis. By analyzing 4,004 rental properties across Tokyo's 23 wards, we developed a multifaceted evaluation approach that combines 2D visibility graph analysis (VGA) from floor plans with 3D architectural element extraction from interior photographs.

Our analysis revealed several significant findings regarding the evolution and distribution of spatial openness in Tokyo's residential architecture:

4.1.1 Temporal Evolution of Spatial Design

The temporal analysis uncovered a clear paradigm shift in residential design philosophy between the 1960s and 1970s. Properties constructed in the 1960s consistently demonstrated higher visibility values across all metrics, with the most pronounced differences observed in the standard deviation of spatial connectivity (effect size $\eta^2 = 0.0609$, medium effect). This suggests that post-war reconstruction era housing featured more spatially integrated and varied layouts compared to the standardized designs that emerged from the 1970s onward.

The progressive compartmentalization of living spaces continued through the 2000s, particularly evident in living room designs where mean visibility values declined by up to 11.72% from the 1960s to 2000s. However, this trend stabilized by the 1990s, indicating that modern residential design has reached an equilibrium between openness and functional separation. Notably, the 2010s marked a partial reversal of this trend through increased window ratios (up to 30.54% higher than previous decades), suggesting a renewed emphasis on natural light and visual connection to the exterior environment.

4.1.2 Geographic Patterns of Openness

The geospatial analysis revealed that spatial openness characteristics are not uniformly distributed across Tokyo but rather follow distinct patterns influenced by urban development and housing typology. Central Tokyo areas demonstrated higher window ratios, likely driven by high-rise apartment developments that prioritize views and natural light. In contrast, suburban areas showed lower visibility values, reflecting traditional Japanese housing preferences for privacy and clearly delineated functional spaces.

The scattered distribution of high visibility properties throughout the city suggests that open floor plans are not concentrated in specific districts but rather represent diverse architectural responses to local conditions, development timing, and target demographics. This finding challenges assumptions about uniform architectural trends within urban zones.

4.1.3 Independence of 2D and 3D Openness Measures

A critical finding emerged from the correlation analysis: 2D spatial connectivity (VGA) and 3D architectural elements (window, ceiling, floor, and wall ratios) operate as independent dimensions of openness. The lack of correlation between these measures indicates that horizontal spatial flow and vertical spatial qualities can be optimized separately, offering designers multiple pathways to create a sense of openness.

This independence was further validated through visual examples showing properties with high visibility but low window ratios achieving openness through layout design, while others with compartmentalized floor plans compensated through extensive glazing. This finding has important implications for design flexibility in constrained urban environments.

4.1.4 Market Valuation of Spatial Openness

The regression analysis demonstrated that spatial openness characteristics contribute moderately to rental price determination ($R^2 = 0.3$). Among the openness indicators, whole-property visibility statistics showed the highest predictive importance, suggesting that overall spatial connectivity matters more to market valuation than room-specific features. The retention of all whole-property visibility indicators (minimum, maximum, mean, and standard deviation) after multicollinearity reduction further emphasizes their distinct contributions to perceived value.

However, the disconnect between objective openness measures and subjective impression scores highlights the complexity of spatial perception. While our quantitative indicators capture important spatial characteristics, factors such as interior design, furniture arrangement, and aesthetic elements play equally important roles in shaping user experiences and preferences.

4.2 Future Work

While this study advances the quantification and analysis of spatial openness in residential environments, several limitations and areas for improvement have been identified that warrant future investigation:

4.2.1 VGA Methodology Refinement

The current VGA implementation faces accuracy limitations due to the granularity constraints of grid-based discretization. Our approach, while scalable, does not achieve the precision of ray-casting visibility algorithms used in specialized architectural software. Future work should explore:

- Development of hybrid approaches that balance computational efficiency with accuracy
- Alignment with established architectural analysis benchmarks to validate results
- Investigation of adaptive grid resolutions that increase density in areas of high spatial complexity
- Integration of more sophisticated visibility algorithms that account for partial occlusions and graduated visual connections

4.2.2 3D Indicator Robustness

The extraction of 3D openness indicators from interior photographs revealed sensitivity to camera angles and shooting positions. Properties photographed from extreme high or low angles showed biased ratios for floor and ceiling elements, potentially leading to misrepresentative conclusions. Future research should address:

- Development of angle-correction algorithms to normalize architectural element ratios
- Direct extraction of lighting conditions using specialized computer vision models
- Integration of depth estimation techniques to better understand spatial volumes
- Creation of standardized photography protocols for rental listings to ensure consistency

4.2.3 Causal Analysis Framework

The current study focused on correlational relationships between openness indicators and various outcomes. However, correlation does not imply causation, and establishing causal links would provide more actionable insights for design and policy decisions. Future work should investigate:

- Development of causal inference frameworks to understand the directional relationships between spatial design choices and outcomes
- Longitudinal studies tracking how renovations affecting openness impact rental prices and tenant satisfaction
- Natural experiments leveraging regulatory changes or development patterns
- Integration of additional confounding variables such as neighborhood characteristics and building amenities

The moderate R^2 values in our regression models suggest that while openness indicators contribute to rental pricing, many other factors remain unaccounted for. A more comprehensive causal framework would help isolate the specific impact of spatial openness from other property attributes.

4.2.4 Scale and Generalizability

The dataset of 4,004 properties, while substantial for a study involving detailed spatial analysis, represents a limited sample given Tokyo's vast rental market. The filtering process necessary to ensure data quality further constrained the sample size. Future research should address:

- Expansion to larger datasets through improved automation of data processing pipelines
- Validation of findings across different cities and cultural contexts
- Development of transfer learning approaches to apply models trained on high-quality data to lower-quality datasets
- Investigation of temporal stability through repeated sampling over multiple years

These future directions would not only address current limitations but also expand the scope and impact of spatial openness research, ultimately contributing to better-designed living environments.

References

[1] B. Hillier, J. Hanson, and H. Graham, "Ideas are in things: An application of the space syntax method to discovering house genotypes," *Environment and Planning B: Planning and design*, vol. 14, no. 4, pp. 363–385, 1987.

[2] A. S. Al-Mohannadi, R. Furlan, and M. D. Major, "A cultural heritage framework for preserving Qatari vernacular domestic architecture," *Sustainability*, vol. 12, no. 18, p. 7295, 2020.

[3] H.-J. Kim and Y.-O. Choi, "Angular VGA and cellular VGA: An exploratory study for spatial analysis methodology based on human movement behavior," p. 054, 2009.

[4] A. Jin, Y. Ge, and S. Zhang, "Spatial characteristics of multidimensional urban vitality and its impact mechanisms by the built environment," *Land*, vol. 13, no. 7, p. 991, 2024.

[5] M. Yamada, X. Wang, and T. Yamasaki, "Graph structure extraction from floor plan images and its application to similar property recommendation," in *2021 IEEE International Conference on Consumer Electronics (ICCE)*, (Las Vegas, NV, USA), pp. 1–2, IEEE, 2021.

[6] A. Turner, "Depthmap: a program to perform visibility graph analysis," in *Proceedings of the 3rd international symposium on space syntax*, vol. 31, pp. 31–12, Citeseer, 2001.

[7] A. Turner, M. Doxa, D. O'Sullivan, and A. Penn, "From isovists to visibility graphs: a methodology for the analysis of architectural space," *Environment and Planning B: Planning and Design*, vol. 28, no. 1, pp. 103–121, 2001.

[8] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," 2018.

[9] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 1290–1299, 2022.

[10] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, "Scene parsing through ade20k dataset," 2017.

[11] T. Oki and T. Shimomura, "Predicting housing preference of households raising children using rental property big data and questionnaire," in *Proceedings of the 30th International Conference of the Association for Computer-Aided Architectural Design Research in Asia (CAADRIA) 2025*, vol. 1, pp. 325–334, 2025.